KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We study slow–fast generators

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}},$$

where γ is the fast relaxation scale and η measures the departure of the effective slow dynamics from detailed balance. Under centered coupling, the slow behaviour is governed at second order by

$$Q_O := -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S.$$

Our main observation is that, in the Dolbeault–Mouhot–Schmeiser method, the corrector should be shifted at the slow spectral gap, namely $\varpi = \lambda(Q_O)$. With this choice we obtain an explicit hypocoercive lower bound at fixed friction. When the stationary state for the chosen parameter pair belongs to the reference Hilbert space, this yields exponential convergence of the semigroup on the mean-zero subspace. For families with uniform structural constants, the bound gives the kinetic scaling

$$\gamma_{\text{opt}} \asymp \max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)},$$

and under additional spectral hypotheses we also prove a complementary upper bound of order $\gamma^{-1} + \eta$ in the overdamped regime. We verify the abstract assumptions for a nonequilibrium Langevin family and for a finite-dimensional two-qubit example.

1. Introduction

1.1. **Problem and main idea.** Many systems of current interest—driven diffusions [15, 22], boundary-driven quantum models [19], and controlled open-system constructions such as reservoir engineering [16]—converge to stationary states that do not satisfy detailed balance. For reversible targets, kinetic lifts can accelerate convergence; this starts with discrete lifting [5] and now appears in diffusion and kinetic settings as square-root improvement of the relaxation scale [2, 10, 1, 20]. The question of the present paper is whether the same mechanism survives when the effective slow dynamics is itself nonequilibrium.

We study generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is the fast relaxation scale, $\mathcal{R} \le 0$ is self-adjoint, $\mathcal{H}_S := \ker(\mathcal{R})$ is the slow space, P_S is the orthogonal projection onto $\mathcal{H}_S$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ measures the departure from detailed balance. Under the centered coupling condition

$$P_S\mathcal{V}P_S = 0, \qquad \mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}},$$

the effective slow operator produced by adiabatic elimination is

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S.$$

Here $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse of $\mathcal{R}$. The corresponding effective operator at $\eta = 0$ is

$$Q_O := -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S. \tag{2}$$

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The spectral gap $\lambda(Q_O)$ is the natural slow scale in the problem.

The main observation is simple. In the Dolbeault–Mouhot–Schmeiser method, the slow contribution of a spectral mode of $-Q_O$ with eigenvalue $\lambda > 0$ is multiplied by

$$\frac{\lambda}{\varpi + \lambda},$$

where $\varpi > 0$ is the shift in the corrector. If ϖ is fixed independently of $\lambda(Q_O)$, then the weakest slow mode is suppressed by an avoidable factor of order $\lambda(Q_O)$. Choosing instead

$$\varpi_* := \lambda(Q_O).$$

keeps the weakest slow mode visible. Section 2 shows this mechanism first in the equilibrium Langevin model, and the later abstract argument is built to preserve it under perturbative nonequilibrium forcing.

1.2. **Main results.** To describe convergence we fix an invariant vector $\mathbf{1} \in \mathcal{H}$ and write

$$\mathcal{H}_0 := \{g \in \mathcal{H} : \langle g, \mathbf{1} \rangle = 0\}.$$

If the parameter pair (γ, η) admits a stationary state $h_{\gamma,\eta} \in \mathbf{1} + \mathcal{H}_0$, then convergence to stationarity is reduced to the semigroup on $\mathcal{H}_0$.

Our first main theorem gives an explicit exponential lower bound on this centered evolution. Under a fast spectral gap, a self-adjoint realization of the effective operator Q_O, a spectral gap $\lambda(Q_O) > 0$ on the mean-zero slow space, a two-term flux estimate, and perturbation bounds, the modified DMS functional with $\varpi = \varpi_*$ yields

$$\|e^{t\mathcal{L}^\dagger} g\| \le C_*(\gamma) e^{-\lambda_\eta t} \|g\|, \qquad g \in \mathcal{H}_0,$$

with $C_*(\gamma) \le 1 + \sqrt{2}$ and an explicit rate lower bound

$$\lambda_\eta \ge (2 - \sqrt{2}) \left(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma) \right).$$

Here $\Lambda_{\mathrm{hypo}}(\gamma)$ is the lower-rate profile produced by the unperturbed 2×2 DMS dissipation matrix, while $B_{\mathrm{pert}}(\gamma)$ is the corresponding linear loss in η.

For model families with uniform structural constants, optimizing the explicit lower bound over γ gives

$$\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}, \qquad \max_{\gamma > 0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}.$$

Thus the certified lower bound has the same square-root scale as the usual kinetic acceleration law.

Our second main theorem is a complementary upper bound. Under additional spectral and contractivity hypotheses, any exponential estimate in the same reference norm is bounded above by a quantity of order $\gamma^{-1} + \eta$. In particular, when $\eta \lesssim \gamma^{-1}$, the overdamped regime remains diffusive. The lower and upper bounds therefore identify intermediate friction as the only place where a genuine acceleration can occur.

We verify the abstract hypotheses in two examples. The nonequilibrium Langevin family is the main unbounded test case, and the two-qubit model keeps the mechanism completely explicit in finite dimensions.

1.3. **Related work.** The modified L^2 hypocoercive framework goes back to Villani [25] and to Dolbeault–Mouhot–Schmeiser [7, 8]. For confining kinetic Fokker–Planck and Langevin equations we also use the regularity viewpoint of Hérau [14]. Explicit rate formulas and spectral discretizations for Langevin dynamics appear in [2, 24], while the periodic nonequilibrium perturbative expansion used as a comparison point below is due to Iacobucci–Olla–Stoltz [15].

From the viewpoint of acceleration by lifting, the starting point is Chen–Lovász–Pak [5]. Recent work on reversible, non-reversible, and second-order lifts includes [10, 11, 1, 9, 20]. The present paper focuses on centered second-order lifts, where the effective operator Q_O is produced by adiabatic elimination and enters directly into the DMS corrector.

For L^2 analysis of nonequilibrium steady states beyond explicit Gibbs weights, relevant references include the stationary-density expansion of [15], Dietert's weighted L^2 framework [6], and Monmarché's recent convergence results for kinetic Langevin dynamics with non-explicit invariant measure [21]; see also [22]. The finite-dimensional example is inspired by dissipative Schrieffer–Wolff and boundary-driven Lindblad reductions [18, 4, 3], but is used here only as an operator-theoretic toy. Quantitative quantum hypocoercive mixing in a different direction is studied in [12].

1.4. **Organization of the paper.** Section 2 presents the equilibrium Langevin warm-up and already shows why the choice $\varpi = \lambda(Q_O)$ is the correct one. Sections 3 and 4 introduce the abstract slow–fast setting, the effective operator on the slow space, and the corresponding modified DMS functional. Section 5 states the abstract lower and upper bounds. Section 6 then discusses a general lift-construction recipe and treats the Langevin and two-qubit examples. The appendices contain the functional-analytic background, the matrix computations, and the model-specific proofs.

2. Warm-up: equilibrium Langevin under a lower Hessian bound

This section isolates the mechanism in the classical equilibrium Langevin model. The choice $\varpi_* = m$ turns the slow DMS coercive term into an order-one contribution, so the lower bound reduces to an explicit 2×2 dissipation matrix estimate. It also shows exactly how nonconvexity enters: assuming only that the position marginal has exact spectral gap $m > 0$ and a lower Hessian bound, the loss is carried by the explicit dual-regularity factor $\sqrt{2 + \frac{K}{2m}}$.

For simplicity we normalize the inverse temperature to 1 in this warm-up. Let

$$d\rho_\infty(x, v) = d\mu(x)\, d\kappa(v), \qquad d\mu(x) = Z^{-1} e^{-U(x)}\, dx, \qquad d\kappa(v) = (2\pi)^{-d/2} e^{-|v|^2/2}\, dv,$$

and write $m > 0$ for the exact spectral gap of the overdamped operator $-\Delta_x + \nabla_x U \cdot \nabla_x$ on $L^2_0(\mu)$, equivalently for the optimal constant in the Poincaré inequality

$$\|g\|^2_{L^2(\mu)} \le \frac{1}{m} \|\nabla_x g\|^2_{L^2(\mu)}, \qquad g \in H^1(\mu), \ \int g\, d\mu = 0, \tag{3}$$

We also assume

$$\nabla^2 U(x) \ge -K \operatorname{Id}, \qquad x \in \mathbb{R}^d, \tag{4}$$

for some $K \ge 0$. Consider on $L^2_0(\rho_\infty)$ the equilibrium Langevin generator

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R}, \qquad \mathcal{V}_{\text{skew}} = v \cdot \nabla_x - \nabla_x U \cdot \nabla_v, \qquad \mathcal{R} = -v \cdot \nabla_v + \Delta_v. \tag{5}$$

The slow projection is the velocity average

$$P_S f(x) := \int_{\mathbb{R}^d} f(x, v)\, d\kappa(v), \qquad P_F := \operatorname{id} - P_S,$$

so the slow space consists of functions of x alone. The associated positive overdamped operator is

$$\mathcal{L}_{\text{ovd}} := -Q_O = -\Delta_x + \nabla_x U \cdot \nabla_x, \tag{6}$$

acting on $L^2_0(\mu)$. By the definition of m through (3), $\mathcal{L}_{\text{ovd}}$ has gap m on $L^2_0(\mu)$, hence $\lambda(Q_O) = m$.

The required confining Kramers–Fokker–Planck regularity is classical [14, Introduction and §§1–2], [8, §2]. The corresponding semigroup and core properties are recalled in Appendix E; they are used only to justify the density and closure step.

The corrector with shift m is

$$\mathcal{A}_m := (m + \mathcal{L}_{\text{ovd}})^{-1} (\mathcal{V}_{\text{skew}} P_S)^\dagger, \tag{7}$$

and the modified Lyapunov functional is

$$\mathsf{E}_\varepsilon(f) := \frac{1}{2} \|f\|^2 - \varepsilon\, \Re \langle \mathcal{A}_m f, f \rangle. \tag{8}$$

Lemma 2.1 (Corrector estimates for the Langevin warm-up)**.** *For every $\phi \in L^2_0(\rho_\infty)$, the corrector $\mathcal{A}_m$ satisfies*

$$\|\mathcal{A}_m\phi\| \le \frac{1}{2\sqrt{m}}\,\|\phi\|, \tag{9}$$

$$\|\mathcal{V}_{\mathrm{skew}}\mathcal{A}_m\phi\| \le \|\phi\|, \tag{10}$$

$$\|\mathcal{A}_m\mathcal{V}_{\mathrm{skew}}P_F\phi\| \le \sqrt{2+\frac{K}{2m}}\,\|\phi\|. \tag{11}$$

Consequently, if $\varepsilon \le \sqrt{m}$, then

$$\frac{1-\varepsilon/\sqrt{m}}{2}\|\phi\|^2 \le \mathsf{E}_\varepsilon(\phi) \le \frac{1+\varepsilon/\sqrt{m}}{2}\|\phi\|^2. \tag{12}$$

Proof. Let $u = \mathcal{A}_m\phi$ and $w = (\mathcal{V}_{\mathrm{skew}}P_S)^\dagger\phi$, so $u = (m+\mathcal{L}_{\mathrm{ovd}})^{-1}w$ and $w \in L^2_0(\mu)$. Since u depends only on x,

$$\|\mathcal{V}_{\mathrm{skew}}u\|^2 = \|\nabla_x u\|^2_{L^2(\mu)} = \langle \mathcal{L}_{\mathrm{ovd}}u, u\rangle_{L^2(\mu)}.$$

Taking the $L^2(\mu)$ inner product of $(m+\mathcal{L}_{\mathrm{ovd}})u = w$ with u and applying Cauchy–Schwarz yields

$$m\|u\|^2 + \|\mathcal{L}^{1/2}_{\mathrm{ovd}}u\|^2 \le \|\phi\|\,\|\mathcal{L}^{1/2}_{\mathrm{ovd}}u\|.$$

This gives both (9) and (10). For (11), set $B := \mathcal{A}_m\mathcal{V}_{\mathrm{skew}}P_F$. If $g \in \mathrm{Ran}(P_S)$ and $h = (m+\mathcal{L}_{\mathrm{ovd}})^{-1}g$, then an integration by parts shows

$$B^\dagger g = -\sum_{i,j}(v_iv_j - \delta_{ij})\,\partial_{ij}h.$$

The Gaussian fourth-moment identity gives $\|B^\dagger g\|^2 = 2\|\nabla^2_x h\|^2_{L^2(\mu)}$. The weighted Bochner identity under (4) yields

$$\|\nabla^2_x h\|^2_{L^2(\mu)} \le \|\mathcal{L}_{\mathrm{ovd}}h\|^2_{L^2(\mu)} + K\|\nabla_x h\|^2_{L^2(\mu)}.$$

Therefore

$$\begin{aligned}\|B^\dagger g\|^2 &\le 2\|\mathcal{L}_{\mathrm{ovd}}h\|^2_{L^2(\mu)} + 2K\langle \mathcal{L}_{\mathrm{ovd}}h, h\rangle_{L^2(\mu)}\\ &= 2\|\mathcal{L}_{\mathrm{ovd}}(m+\mathcal{L}_{\mathrm{ovd}})^{-1}g\|^2_{L^2(\mu)} + 2K\left\langle \mathcal{L}_{\mathrm{ovd}}(m+\mathcal{L}_{\mathrm{ovd}})^{-2}g, g\right\rangle_{L^2(\mu)}\\ &\le \Big(2 + 2K\sup_{\lambda\ge 0}\frac{\lambda}{(m+\lambda)^2}\Big)\|g\|^2_{L^2(\mu)} = \Big(2+\frac{K}{2m}\Big)\|g\|^2_{L^2(\mu)}.\end{aligned}$$

Hence $\|B^\dagger g\| \le \sqrt{2+\frac{K}{2m}}\,\|g\|$, which is (11). The norm equivalence follows from (9) and Cauchy–Schwarz. □

Write $g_S = P_S g$ and $g_F = P_F g$. The Gaussian Poincaré inequality in v gives $\|g_F\|^2 \le \|\nabla_v g\|^2$. Therefore, if we set

$$X := \|\nabla_v g\|, \qquad Y := \|g_S\|, \qquad Z := \frac{\gamma}{2\sqrt{m}} + \sqrt{2+\frac{K}{2m}},$$

then differentiating E_ε along $\partial_t g = \mathcal{L}g$ and using Lemma 2.1 yields

$$\frac{d}{dt}\mathsf{E}_\varepsilon(g) \le -\begin{pmatrix}X\\ Y\end{pmatrix}^T\begin{pmatrix}\gamma-\varepsilon & -\frac{\varepsilon Z}{2}\\ -\frac{\varepsilon Z}{2} & \frac{\varepsilon}{2}\end{pmatrix}\begin{pmatrix}X\\ Y\end{pmatrix}. \tag{13}$$

Indeed, the microscopic dissipation gives $-\gamma X^2$; the macroscopic term is

$$-\varepsilon\Big\langle \frac{\mathcal{L}_{\mathrm{ovd}}}{m+\mathcal{L}_{\mathrm{ovd}}}g_S, g_S\Big\rangle \le -\frac{\varepsilon}{2}Y^2$$

because $\mathcal{L}_{\mathrm{ovd}} \geq m$ on $L^2_0(\mu)$; the fast–fast remainder contributes at most εX^2 by (10); the dual regularity estimate (11) contributes $\varepsilon\sqrt{2+\frac{K}{2m}}\,XY$; and the correction involving $\gamma\mathcal{R}$ contributes $\varepsilon\gamma(2\sqrt{m})^{-1}XY$ by (9).

Proposition 2.2. *For the corrector $\mathcal{A}_m$, the DMS lower-rate profile (that is, the smallest eigenvalue of the dissipation matrix before converting the Lyapunov functional back to the L^2 norm) satisfies, for every $\gamma > 0$,*

$$\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma) \geq \frac{\gamma}{4\Big(\big(\gamma/(2\sqrt{m}) + \sqrt{2+\frac{K}{2m}}\big)^2 + 2\Big)}. \tag{14}$$

The displayed lower bound is maximized at $\gamma_{\mathrm{opt}} = \sqrt{16m+2K}$, yielding

$$\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}}) \geq \frac{\sqrt{m}}{4\Big(\sqrt{2+\frac{K}{2m}} + \sqrt{4+\frac{K}{2m}}\Big)}. \tag{15}$$

Since $m = \lambda(Q_O)$, this is exactly the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$ whenever $K = \mathcal{O}(m)$; when $K = 0$, (15) reduces to $\frac{2-\sqrt{2}}{8}\sqrt{m}$.

Proof. Let $M_{\gamma,\varepsilon}$ denote the dissipation matrix in (13). Its determinant is

$$\det M_{\gamma,\varepsilon} = \frac{\gamma\varepsilon}{2} - \frac{\varepsilon^2(Z^2+2)}{4},$$

so the natural choice is

$$\varepsilon_* = \frac{\gamma}{Z^2+2}.$$

For a positive 2×2 matrix, $\lambda_{\min}(M) \geq \det(M)/\operatorname{tr}(M)$, and here $\operatorname{tr}(M_{\gamma,\varepsilon_*}) = \gamma - \varepsilon_*/2 \leq \gamma$. Hence

$$\lambda_{\min}(M_{\gamma,\varepsilon_*}) \geq \frac{\gamma}{4(Z^2+2)}.$$

This gives the stated lower bound. Writing $x = \gamma/(2\sqrt{m})$, the right-hand side becomes

$$\sqrt{m}\,\frac{x}{2\Big(\big(x+\sqrt{2+\frac{K}{2m}}\big)^2+2\Big)}.$$

A direct derivative computation shows that it is maximized at $x = \sqrt{4+\frac{K}{2m}}$, i.e. at $\gamma_{\mathrm{opt}} = \sqrt{16m+2K}$. Substituting this value yields (15). □

Corollary 2.3 (Warm-up semigroup rate at the optimizing friction)**.** *At this maximizing friction, set*

$$\varepsilon_* := \frac{\gamma_{\mathrm{opt}}}{Z(\gamma_{\mathrm{opt}})^2+2}.$$

Then the semigroup generated by $\mathcal{L}$ satisfies

$$\|e^{t\mathcal{L}}g\|_{L^2(\rho_\infty)} \leq \sqrt{3}\,\exp\left(-\frac{\sqrt{m}}{6\Big(\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}\Big)}\,t\right)\|g\|_{L^2(\rho_\infty)}, \qquad g\in L^2_0(\rho_\infty),\ t\geq 0. \tag{16}$$

In particular, when $K = 0$ the decay rate is $\frac{2-\sqrt{2}}{12}\sqrt{m}$.

Proof. At $\gamma = \gamma_{\mathrm{opt}}$, Proposition 2.2 gives the matrix-profile bound

$$\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}}) \geq \frac{\sqrt{m}}{4\Big(\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}\Big)}. \tag{17}$$

For the associated optimizer ε_* one has

$$\varepsilon_* = \frac{\sqrt{m}}{\sqrt{2+\frac{K}{2m}}+\sqrt{4+\frac{K}{2m}}} \le \frac{\sqrt{m}}{2}, \tag{18}$$

so Lemma 2.1 yields

$$\frac{1}{4}\|g\|^2 \le \mathsf{E}_{\varepsilon_*}(g) \le \frac{3}{4}\|g\|^2. \tag{19}$$

The dissipation estimate (13) with this optimal choice gives

$$\frac{d}{dt}\mathsf{E}_{\varepsilon_*}(g_t) \le -\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}})\,\|g_t\|^2 \le -\frac{4}{3}\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}})\,\mathsf{E}_{\varepsilon_*}(g_t). \tag{20}$$

Gronwall's lemma then gives

$$\|g_t\|^2 \le 3\,e^{-\frac{4}{3}\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}(\gamma_{\mathrm{opt}})t}\,\|g_0\|^2, \tag{21}$$

which is exactly the stated semigroup estimate after taking square roots. □

Warm-up profile versus semigroup rate. Proposition 2.2 records the matrix-profile constant $\Lambda^{\mathrm{Lan}}_{\mathrm{hypo}}$, while Corollary 2.3 gives the corresponding semigroup rate. The latter differs from the former by the standard Lyapunov-to-L^2 conversion factor $2/3$.

For comparison, any fixed shift independent of m leaves the slow coercive block of order m and therefore fails to capture the kinetic-scale lower bound as $m \downarrow 0$. Thus the warm-up already contains the mechanism behind the abstract theory: the choice $\varpi_* = m$ restores the kinetic scale whenever m tracks the slow gap, while a lower Hessian bound only changes the explicit dual-regularity constant. The later sections ask how much of this survives after the perturbation $\eta\mathcal{V}_{\mathrm{pert}}$ is switched on.

3. Preliminaries

We now formulate the abstract slow–fast setting used in the main theorem. Section 2 should be kept in mind as the model case: the role of the scalar gap m will be played by the slow spectral gap $\lambda(Q_O)$, and the concrete Langevin corrector will become an abstract resolvent on the slow space.

3.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a separable complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ (conjugate-linear in the first entry) and norm $\|\cdot\|$. We fix a distinguished unit vector $\mathbf{1} \in \mathcal{H}$ and set

$$\mathcal{H}_0 := \{u \in \mathcal{H} : \langle u, \mathbf{1}\rangle = 0\}, \qquad \mathcal{H}_1 := \mathbf{1} + \mathcal{H}_0.$$

The generator $\mathcal{L}$ is viewed on this fixed reference space, and $\mathcal{L}^\dagger$ denotes its Hilbert adjoint. In the classical setting $\mathcal{H} = L^2(\mu)$ and $\mathbf{1}$ is the constant-one function; in the quantum setting $\mathcal{H} = L^2(\sigma)$ with KMS inner product

$$\langle X, Y\rangle_\sigma := \mathrm{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y),$$

and $\mathbf{1}$ is the identity observable. Whenever $\mathcal{L}\mathbf{1} = 0$, the subspace $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$. If, for a given parameter pair (γ,η), there exists a stationary vector/state $h_{\gamma,\eta} \in \mathcal{H}_1$ with $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$, then every $f \in \mathcal{H}_1$ determines a centered datum $g := f - h_{\gamma,\eta} \in \mathcal{H}_0$. All quantitative estimates below are stated on $\mathcal{H}_0$; Lemma 4.2 records the passage back to physical densities, states, or normalized stationary vectors.

Dense-core convention in the unbounded setting. In finite dimensions all operators are bounded. In unbounded settings we fix a dense invariant core

$$\mathcal{C} \subset \mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathrm{dom}((-\mathcal{R})^{1/2})$$

on which $\mathcal{R}$, $\mathcal{V}_{\mathrm{skew}}$, $\mathcal{V}_{\mathrm{pert}}$, P_S, P_F, and $\mathcal{R}^+$ preserve $\mathcal{C}$, the adjoint identity

$$\langle \mathcal{V}_{\mathrm{skew}} u, v\rangle = -\langle u, \mathcal{V}_{\mathrm{skew}} v\rangle$$

holds for $u, v \in \mathcal{C}$, and the compositions defining Q_O, $\mathcal{A}_\varpi$, and the differentiated Lyapunov identity are well defined; see [7, 8, 25]. For every $u_0 \in \mathrm{dom}(\mathcal{L}^\dagger)$, the orbit $u(t) = e^{t\mathcal{L}^\dagger} u_0$ belongs to $C([0,\infty); \mathrm{dom}(\mathcal{L}^\dagger)) \cap C^1([0,\infty); \mathcal{H})$ and satisfies $\partial_t u = \mathcal{L}^\dagger u$. All differentiated identities are first proved on $\mathcal{C}$; the resulting dissipation forms are continuous for the graph norm $\|u\| + \|\mathcal{L}^\dagger u\|$, so they extend from $\mathcal{C}$ to $\mathrm{dom}(\mathcal{L}^\dagger)$ by the core property, and the final semigroup estimates pass from $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ to $\mathcal{H}_0$ by strong continuity of $e^{t\mathcal{L}^\dagger}$. For the Langevin family we take $\mathcal{C} = \mathscr{S}(\mathbb{R}^{2d})$; see Appendix E.

3.2. **Spectral gaps.** Write

$$\mathcal{L}_{\mathrm{sym}} := \tfrac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger)$$

for the symmetric part of the generator. Whenever $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$, we denote by $\mathcal{L}_0^\dagger$ the generator of the restricted semigroup on $\mathcal{H}_0$.

Definition 3.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap*

$$\lambda_S := \inf_{f \in \mathcal{H}_0 \setminus \{0\}} \frac{-\langle f, \mathcal{L}_{\mathrm{sym}} f\rangle}{\|f\|^2},$$

and the *real-part spectral gap*

$$\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \mathrm{Spec}(-\mathcal{L}_0^\dagger)\},$$

whenever this restricted generator is defined. Since $\mathrm{Spec}(\mathcal{L}^\dagger) = \overline{\mathrm{Spec}(\mathcal{L})}$, the real-part gap is the same for forward and backward dynamics.

When $\mathcal{L}$ is self-adjoint and non-positive on $\mathcal{H}_0$ (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, λ_S need not control relaxation and may vanish while the real-part spectral gap $\lambda(\mathcal{L})$ is positive: the skew part $\frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [25, 7]. This quantity is a spectral abscissa; when the sharp semigroup estimate requires a prefactor $C > 1$, it need not coincide with the optimal decay rate in norm. In the present reference-space setting the basic estimate takes the form

$$\|e^{t\mathcal{L}^\dagger} g\| \le C e^{-\lambda t} \|g\|, \qquad g \in \mathcal{H}_0,$$

and when $C > 1$ is necessary we refer to the relaxation as *strictly hypocoercive*.

3.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \le 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = f_S + f_F$ with $f_S := P_S f$ (slow/macroscopic) and $f_F := P_F f$ (fast/microscopic). For the classical and bounded operator families considered below, the reference stationary mode satisfies $\mathbf{1} \in \mathcal{H}_S$. Therefore,

$$\mathcal{H}_F \subset \mathcal{H}_0, \qquad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

Hence, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.2 (Centered coupling)**.** The interaction satisfies $P_S \mathcal{V} P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}, generated by slow–fast–slow excursions. Without it one already sees an $\mathcal{O}(1)$ tangential drift at the block-operator level.

Definition 3.3 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{22}$$

The operator $\mathcal{L}_O$ is the effective slow dynamics produced by adiabatic elimination. In settings where singular perturbation/adiabatic elimination is justified, one expects the macroscopic-time evolution to be governed by $e^{\tau \mathcal{L}_O}$ with $\tau = t/\gamma$; see, for instance, [3] in a specific boundary-driven regime. This is the origin of the diffusive γ^{-1} bottleneck behind the upper bound proved later.

When $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$, we impose the stronger componentwise centering conditions

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0.$$

This yields estimates uniform in η and isolates the effective operator Q_O at $\eta = 0$.

3.4. **Analytic tools for the slow–fast operators.** The appendix records one sufficient route to the slow operator Q_O: the spectral-calculus construction of $\mathcal{R}^+$ and the Friedrichs construction of the slow quadratic form. In the abstract theory, however, we do not derive this realization from the standing hypotheses; instead it is assumed in Assumption 4.4 and verified for the Langevin family in Appendix E. In finite dimensions every step is trivial. We record only the key consequences used later:

(i) $\mathcal{R}^+$ is bounded and self-adjoint with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$ and $\|(-\mathcal{R}^+)^{1/2}\| \le \lambda_R^{-1/2}$;
(ii) Q_O is non-positive and self-adjoint on $\mathcal{H}_S$, with the representation properties recorded in Assumption 4.4;
(iii) for every $\varpi > 0$, $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$, and $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \le (\varpi + \lambda(Q_O))^{-1}$ when $\lambda(Q_O) > 0$.

4. Abstract framework

Section 2 showed the mechanism in one concrete model. We now record the abstract assumptions used in the rest of the paper. The point is not to introduce a second theory, but to isolate exactly which structural inputs from the warm-up are needed later: a fast spectral gap, centered coupling, control of the induced slow–fast flux, and perturbation estimates.

Assumption 4.1 (Mass conservation and a stationary state)**.** The distinguished vector $\mathbf{1}$ belongs to $\mathcal{H}_S \cap \mathrm{dom}(\mathcal{L})$ and satisfies

$$\mathcal{L}\mathbf{1} = 0.$$

Hence the forward semigroup $e^{t\mathcal{L}^\dagger}$ preserves the reference mass:

$$\langle e^{t\mathcal{L}^\dagger} f, \mathbf{1} \rangle = \langle f, \mathbf{1} \rangle, \qquad t \ge 0.$$

In particular, the codimension-one subspace $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$. For the fixed parameter pair (γ, η) under consideration, there exists a stationary vector/state $h_{\gamma,\eta} \in \mathcal{H}_1 \cap \mathrm{dom}(\mathcal{L}^\dagger)$ such that

$$\mathcal{L}^\dagger h_{\gamma,\eta} = 0.$$

This is the only place where the stationary state enters the abstract argument; all coercive estimates below are carried out on $\mathcal{H}_0$. The remaining assumptions describe the effective operator Q_O on the slow space and the bounds needed to control the DMS cross term. Theorem 5.1 applies whenever both sets of assumptions hold for the parameter pair under consideration.

Lemma 4.2 (Centered reduction)**.** *Assume Assumption 4.1. Let $f \in \mathcal{H}_1$ and $f_t := e^{t\mathcal{L}^\dagger} f$. If*

$$g := f - h_{\gamma,\eta}, \qquad g_t := f_t - h_{\gamma,\eta},$$

then $g \in \mathcal{H}_0$ and

$$g_t = e^{t\mathcal{L}^\dagger} g \in \mathcal{H}_0.$$

If, in addition, $f \in \mathrm{dom}(\mathcal{L}^\dagger)$, then $\partial_t g_t = \mathcal{L}^\dagger g_t$. Consequently, any semigroup estimate proved on $\mathcal{H}_0$ transfers verbatim to convergence of f_t toward $h_{\gamma,\eta}$.

Proof. Since $\langle f, \mathbf{1}\rangle = \langle h_{\gamma,\eta}, \mathbf{1}\rangle = 1$, one has $g \in \mathcal{H}_0$. Because $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$ and $e^{t\mathcal{L}^\dagger} h_{\gamma,\eta} = h_{\gamma,\eta}$,

$$g_t = e^{t\mathcal{L}^\dagger} f - h_{\gamma,\eta} = e^{t\mathcal{L}^\dagger}(f - h_{\gamma,\eta}) = e^{t\mathcal{L}^\dagger} g.$$

Mass conservation gives $\langle g_t, \mathbf{1}\rangle = 0$, so $g_t \in \mathcal{H}_0$. If $f \in \mathrm{dom}(\mathcal{L}^\dagger)$, then also $g \in \mathrm{dom}(\mathcal{L}^\dagger)$ and the evolution equation follows from the standard differentiability of the semigroup on the operator domain. □

Lemma 4.3 (Centered part generator)**.** *Assume Assumption 4.1. Then $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ is dense in $\mathcal{H}_0$. Moreover, the restriction of $e^{t\mathcal{L}^\dagger}$ to $\mathcal{H}_0$ is a C_0-semigroup. Its generator is the part of $\mathcal{L}^\dagger$ in $\mathcal{H}_0$, namely*

$$\mathcal{L}_0^\dagger g := \mathcal{L}^\dagger g, \qquad \mathrm{dom}(\mathcal{L}_0^\dagger) = \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0.$$

Proof. Let $g \in \mathcal{H}_0$. Since $\mathrm{dom}(\mathcal{L}^\dagger)$ is dense in $\mathcal{H}$, choose $f_n \in \mathrm{dom}(\mathcal{L}^\dagger)$ with $f_n \to g$ in $\mathcal{H}$. Set

$$g_n := f_n - \langle f_n, \mathbf{1}\rangle\, h_{\gamma,\eta}.$$

Because $h_{\gamma,\eta} \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_1$, one has $g_n \in \mathrm{dom}(\mathcal{L}^\dagger)$ and

$$\langle g_n, \mathbf{1}\rangle = \langle f_n, \mathbf{1}\rangle - \langle f_n, \mathbf{1}\rangle\langle h_{\gamma,\eta}, \mathbf{1}\rangle = 0,$$

so $g_n \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. Since $f_n \to g$ and $\langle f_n, \mathbf{1}\rangle \to \langle g, \mathbf{1}\rangle = 0$, we also have $g_n \to g$ in $\mathcal{H}$. This proves density.

By Assumption 4.1, $\mathcal{H}_0$ is invariant under $e^{t\mathcal{L}^\dagger}$, so the restriction is a C_0-semigroup on the closed subspace $\mathcal{H}_0$. By the standard restriction theorem for invariant subspaces, its generator is the part of $\mathcal{L}^\dagger$ in $\mathcal{H}_0$ [23, Thm. 1.2.5]. If $g \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$, then

$$\langle \mathcal{L}^\dagger g, \mathbf{1}\rangle = \langle g, \mathcal{L}\mathbf{1}\rangle = 0$$

because $\mathbf{1} \in \mathrm{dom}(\mathcal{L})$ and $\mathcal{L}\mathbf{1} = 0$. Hence $\mathcal{L}^\dagger g \in \mathcal{H}_0$, and the part-domain is exactly $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$. □

4.1. **Abstract assumptions.** Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \geq 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{23}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

The body estimates depend on three ingredients beyond mass conservation: a non-positive self-adjoint effective operator Q_O on $\mathcal{H}_S$, quantitative control of the fast pseudoinverse and the induced microscopic fluxes, and perturbation bounds. In unbounded models one sufficient route to Q_O is the Friedrichs construction from Appendix A; in the abstract theory we simply assume the realization actually used in the proof.

Assumption 4.4 (Self-adjoint effective operator on the slow space)**.** There exists a non-positive self-adjoint operator Q_O on $\mathcal{H}_S$ and a dense subspace $\mathcal{C}_Q \subset \mathrm{dom}(Q_O) \cap \mathrm{dom}(\mathcal{V}_{\mathrm{skew}}) \cap \mathcal{H}_S$, containing the slow slice of the common differentiation core ($P_S\mathcal{C} \subset \mathcal{C}_Q$), such that:

(i) for every $\psi \in \mathcal{C}_Q$,

$$Q_O\psi = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S\psi; \tag{24}$$

(ii) for every $\psi \in \text{dom}(Q_O)$, the microscopic flux

$$w_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi$$

is well defined in $\mathcal{H}_F$ and satisfies

$$\langle \psi, (-Q_O)\psi\rangle = \langle w_\psi, (-\mathcal{R}^+)w_\psi\rangle; \tag{25}$$

(iii) for every $\psi \in \text{dom}(Q_O)$, the slow image $P_S\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi$ is well defined and obeys

$$P_S\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi = Q_O\psi. \tag{26}$$

The assumptions below encode four additional inputs only: a spectral gap on the fast space, centered slow–fast transport, a closure estimate for the induced microscopic flux, and perturbation bounds.

Assumption 4.5 (Structural conditions)**.** The generator has the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The standing dense-core convention from Section 3.1 remains in force; in particular, the skew-adjointness identity for $\mathcal{V}_{\text{skew}}$ on the common core is part of the abstract setup. We assume:

(1) **Microscopic spectral gap.** $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$, and there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \geq \lambda_R\|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{27}$$

(2) **Centered coupling.**

$$P_S\mathcal{V}_{\text{skew}}P_S = 0, \qquad P_S\mathcal{V}_{\text{pert}}P_S = 0. \tag{28}$$

(3) **Flux regularity.** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$, let w_ψ be the microscopic flux from Assumption 4.4(ii). There exist constants $M_R, \kappa_{\text{ell}}, \kappa_{\text{low}} > 0$ such that

$$\|w_\psi\|^2 \leq M_R\,\langle w_\psi, -\mathcal{R}^+w_\psi\rangle, \qquad \forall\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{29}$$

$$\|P_F\mathcal{V}_{\text{skew}}P_F\,(-\mathcal{R}^+)\,w_\psi\| \leq \ \kappa_{\text{ell}}\|(-Q_O)\psi\| + \kappa_{\text{low}}\,\|(-Q_O)^{1/2}\psi\|. \tag{30}$$

(4) **Structural decomposition of the perturbation.**

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{31}$$

where

$$\mathcal{K}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger), \qquad \mathcal{S}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger),$$

with the adjoint taken in $\mathcal{H}$. Assume:

(a) **Skew perturbation control.** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric on its domain. In particular, for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}}f, f\rangle = 0. \tag{32}$$

There exist constants $a_1, b_1 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}}f\| \leq a_1\,\|(-\mathcal{R})^{1/2}f\| + b_1\,\|f\|. \tag{33}$$

(b) **Symmetric perturbation control.** $\mathcal{S}_{\text{pert}}$ is symmetric on $\text{dom}((-\mathcal{R})^{1/2})$, and there exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}}f\| \leq a_2\,\|(-\mathcal{R})^{1/2}f\| + b_2\,\|f\|. \tag{34}$$

Assumption 4.4 identifies the effective operator on the slow space. Item (iii) of Assumption 4.5 is the abstract analogue of the weighted elliptic estimate in the Langevin warm-up, while item (iv) records the price of nonequilibrium forcing. The split in (30) is intentional: κ_{ell} controls the top-order term, whereas κ_{low} records lower-order defects such as the Hessian contribution in the Langevin example. The final ingredient is a spectral gap for Q_O.

Assumption 4.6 isolates the positive spectral gap needed on the slow space.

Assumption 4.6 (Spectral gap of the effective operator)**.** Because $\mathbf{1} \in \mathcal{H}_S$, we write $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$. We assume that

$$\mathbf{1} \in \mathrm{dom}(Q_O), \qquad Q_O\mathbf{1} = 0,$$

and that there exists a constant $\lambda(Q_O) > 0$ such that

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f\|^2, \qquad \forall f \in \mathrm{dom}(Q_O) \cap \mathcal{H}_{S,0}. \tag{35}$$

Since Q_O is self-adjoint and $Q_O\mathbf{1} = 0$, the subspace $\mathcal{H}_{S,0} = \mathbf{1}^\perp \cap \mathcal{H}_S$ is Q_O-invariant. Equivalently, the restricted self-adjoint operator $-Q_O|_{\mathcal{H}_{S,0}}$ has spectrum contained in $[\lambda(Q_O), \infty)$. Consequently, $\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S$, and $(\varpi - Q_O)^{-1}$ is bounded on $\mathcal{H}_S$ for every $\varpi > 0$.

No further abstract domain input is used beyond Assumption 4.4 and the standing dense-core convention from Section 3.1: in the unbounded case the Lyapunov differentiation is carried out first on the common core, whose slow slice lies in $\mathcal{C}_Q$, and then extended by graph-norm continuity. This is verified for the Langevin family in Appendix E.

4.2. **The corrector.** The corrector is the abstract version of the resolvent used in the warm-up. It solves a shifted equation on the slow space and produces the cross term in the modified energy functional.

Lemma 4.7 (Resolvent of the effective operator)**.** *Let Q_O be as in Assumptions 4.4 and 4.6. Then for every shift parameter $\varpi > 0$,*

- *(i) the operator $(\varpi - Q_O)$ is strictly positive on $\mathcal{H}_S$ with spectrum contained in $[\varpi, \infty)$;*
- *(ii) the resolvent $(\varpi - Q_O)^{-1}$ is a bounded, self-adjoint, positive operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \leq \varpi^{-1}$;*
- *(iii) on the mean-zero subspace $\mathcal{H}_{S,0}$, the sharper bound $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \leq (\varpi + \lambda(Q_O))^{-1}$ holds;*
- *(iv) $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$.*

Proof. Since $-Q_O \geq 0$ is self-adjoint and $Q_O\mathbf{1} = 0$, the subspace $\mathcal{H}_{S,0} = \mathbf{1}^\perp \cap \mathcal{H}_S$ is Q_O-invariant. Assumption 4.6 therefore gives

$$\sigma(-Q_O|_{\mathcal{H}_{S,0}}) \subset [\lambda(Q_O), \infty), \qquad \sigma(-Q_O|_{\mathbb{C}\mathbf{1}}) = \{0\}.$$

Therefore $\sigma(\varpi - Q_O) \subset [\varpi, \varpi + \|Q_O\|]$ in finite dimensions, or $\sigma(\varpi - Q_O) \subset [\varpi, \infty)$ in general. Items (i)–(iii) are direct consequences; item (iv) is the standard resolvent mapping property of the closed self-adjoint operator Q_O. □

Definition 4.8 (Corrector)**.** Let $\varpi > 0$ be a shift parameter. In Section 5 we will choose $\varpi_* = \lambda(Q_O)$. On the common core from Section 3.1, we define the **corrector** by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F. \tag{36}$$

Lemma 4.9 shows that this core operator extends uniquely to a bounded operator $\mathcal{H} \to \mathcal{H}_S$; we keep the same notation for the extension.

Because dissipation vanishes on $\mathcal{H}_S$, the plain energy does not see the slow modes. The role of $\mathcal{A}_\varpi$ is to transfer coercivity of $-Q_O$ into a bounded cross term. On a slow spectral mode of $-Q_O$ with eigenvalue $\lambda > 0$, the main DMS contribution carries the factor $\lambda/(\varpi + \lambda)$. The choice $\varpi = \lambda(Q_O)$ therefore guarantees that every mode in $\mathcal{H}_{S,0}$ contributes at least $1/2$.

Lemma 4.9 (Structural Estimates for the DMS Corrector)**.** *Under Assumptions 4.4 and 4.5, the core operator $\mathcal{A}_\varpi$ extends uniquely to a bounded operator $\mathcal{H} \to \mathcal{H}_S$, still denoted by $\mathcal{A}_\varpi$. This extension satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, \kappa_{\text{low}}, a_1, b_1, a_2, b_2)$ and with the only shift dependence displayed explicitly below:*

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}} \varpi^{-1/2}, \quad \text{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{37}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \le k_{\text{reg}}, \quad \text{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{38a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \le k_{\text{dual}}(\varpi), \quad \text{with } k_{\text{dual}}(\varpi) := \kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}}. \tag{38b}$$

The dual estimate (38b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 4.5.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in all reference-space directions:*

$$\|\mathcal{V}_{\text{pert}} \mathcal{A}_\varpi\|, \ \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi\| \le k_P(\varpi), \tag{39a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}\|, \ \|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger\| \le \tilde{k}_P(\varpi). \tag{39b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}}\, \varpi^{-1/2}, \tag{40}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2}. \tag{41}$$

Moreover, the centered perturbation coupling in Assumption 4.5.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^\dagger P_S = 0, \tag{42}$$

and therefore

$$\mathcal{V}_{\text{pert}}\, \mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^\dagger\, \mathcal{H}_S \subset \mathcal{H}_F. \tag{43}$$

Proof. See Appendix B. □

4.3. **The modified Lyapunov functional.**

Definition 4.10 (Modified Lyapunov functional [7, 8])**.** Because the quantitative proof is carried out for the forward density semigroup $e^{t\mathcal{L}^\dagger}$, the DMS cross term enters with the sign adapted to the adjoint dynamics. For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 - \alpha \Re\langle \mathcal{A}_\varpi f, f\rangle. \tag{44}$$

This functional defines a norm equivalent to the Hilbert norm once $\alpha\|\mathcal{A}_\varpi\|$ is small.

Proposition 4.11 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| < 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{45}$$

Proof. Immediate from $|\Re\langle \mathcal{A}_\varpi f, f\rangle| \le \|\mathcal{A}_\varpi\|\, \|f\|^2$ and the definition of $\mathcal{E}_{\alpha,\varpi}$. □

5. Main abstract results

We now state the two abstract results: an explicit lower bound and a complementary upper bound. Together they identify the friction regime in which acceleration can occur.

5.1. **Explicit lower bound.** We first fix $\varpi_* = \lambda(Q_O)$ and optimize only the Lyapunov weight α. As in the warm-up, the proof then reduces to a 2×2 dissipation matrix. The matrix computation is deferred to Appendix C; here we record the resulting explicit constants.
Explicit constants. We fix $\varpi_* := \lambda(Q_O)$. Three quantities govern the theorem: the coupling scale $Z(\gamma)$, the unperturbed lower-rate profile $\Lambda_{\mathrm{hypo}}(\gamma)$, and the perturbative loss $B_{\mathrm{pert}}(\gamma)$. To write them compactly, set

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\mathrm{geom}} := \kappa_{\mathrm{ell}} + \frac{\kappa_{\mathrm{low}}}{2\sqrt{\lambda(Q_O)}}, \qquad b_{2,\mathrm{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{46}$$

Here Θ measures the size of the slow–fast flux, and $K_{\mathrm{geom}} = k_{\mathrm{dual}}(\lambda(Q_O))$ is the effective transport constant at the chosen shift.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}} = \kappa_{\mathrm{ell}} + \big(\gamma\Theta + \tfrac{1}{2}\kappa_{\mathrm{low}}\big)\lambda(Q_O)^{-1/2}, \tag{47}$$

which measures the strength of slow–fast interaction at friction γ. We then set

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}, \tag{48}$$

$$\Lambda_{\mathrm{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}, \tag{49}$$

$$B_{\mathrm{pert}}(\gamma) := b_{2,\mathrm{eff}} + \alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2}, \tag{50}$$

where $C_{\mathrm{sing}} := 2(b_1+b_2)\, k_{\mathcal{A}} + (a_1+a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.9. Thus $\alpha_*(\gamma)$ is the optimized Lyapunov weight, $\Lambda_{\mathrm{hypo}}(\gamma)$ is the unperturbed lower-rate profile, and $B_{\mathrm{pert}}(\gamma)$ is the $\mathcal{O}(\eta)$ loss. The apparent singular factor $\lambda(Q_O)^{-1/2}$ in B_{pert} is cancelled at optimal friction; see Corollary 5.2.

We also isolate the explicit admissibility restriction coming from the symmetric perturbation in the basic energy estimate:

$$\eta_{\mathrm{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0. \end{cases} \tag{51}$$

This is simply the condition under which the coefficient of $\|(-\mathcal{R})^{1/2} f\|^2$ stays nonnegative in the basic energy estimate; see Appendix C.

Finally, define the Lyapunov-distortion parameter and norm-equivalence prefactor by

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|, \tag{52}$$

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}. \tag{53}$$

The appendix proves the uniform bound $\delta_*(\gamma) \le 2^{-1/2}$ for all $\gamma > 0$. Hence Proposition 4.11 applies to the optimized Lyapunov functional and $C_*(\gamma) \le 1 + \sqrt{2}$.

The threshold $\eta^*(\gamma)$ defined below controls only the coercivity argument. In concrete unbounded models it must be combined with an independent condition guaranteeing that the stationary state $h_{\gamma,\eta}$ belongs to the chosen reference space.

Theorem 5.1 (Explicit exponential lower bound)**.** *Let $\varpi_* := \lambda(Q_O) > 0$ be the slow-gap scale from Assumption 4.6, and set the explicit certified threshold*

$$\eta^*(\gamma) := \min\left\{\eta_{\mathrm{sym}}(\gamma),\ \frac{\Lambda_{\mathrm{hypo}}(\gamma)}{2B_{\mathrm{pert}}(\gamma)}\right\}.$$

Fix $\gamma > 0$ and $0 \le \eta \le \eta^*(\gamma)$. Suppose Assumptions 4.1, 4.4, 4.5, and 4.6 hold for this parameter pair. Then the semigroup on $\mathcal{H}_0$ satisfies

$$\|e^{t\mathcal{L}^\dagger} g\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|g\| \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|g\|, \quad \forall g \in \mathcal{H}_0,\ \forall t \ge 0. \tag{54}$$

The decay rate satisfies

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\left(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\right). \tag{55}$$

By Lemma 4.2, the same bound yields exponential convergence of $f_t = e^{t\mathcal{L}^\dagger} f$ toward $h_{\gamma,\eta}$ for every $f \in \mathcal{H}_1$.

Proof. By Lemma 4.2, it suffices to work on $\mathcal{H}_0$. Let $g_t = e^{t\mathcal{L}^\dagger} g$ with $g \in \mathcal{H}_0$. Differentiating the Lyapunov functional $\mathcal{E}_{\alpha,\varpi_*}(g_t)$ on the core gives a 2×2 dissipation matrix in the fast and slow components. The choice $\varpi_* = \lambda(Q_O)$ makes the slow diagonal entry order one, while Assumption 4.5.4 contributes only the explicit $\mathcal{O}(\eta)$ correction encoded in $B_{\mathrm{pert}}(\gamma)$.

Choosing $\alpha = \alpha_*(\gamma)$ optimizes the determinant of this matrix and yields the profile $\Lambda_{\mathrm{hypo}}(\gamma)$. The condition $\eta \le \eta^*(\gamma)$ guarantees positivity of the optimized matrix, so

$$\frac{d}{dt}\mathcal{E}_{\alpha_*(\gamma),\varpi_*}(g_t) \le -\big(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\big)\|g_t\|^2.$$

The uniform bound $\delta_*(\gamma) \le 2^{-1/2}$ and Proposition 4.11 convert this differential inequality into the stated semigroup estimate on $\mathcal{H}_0$. The full matrix computation, the graph-norm closure argument, and the passage from the core to the full semigroup are given in Appendix C. □

Thus Theorem 5.1 should be read pointwise in γ: $\Lambda_{\mathrm{hypo}}(\gamma)$ is the explicit lower-rate profile for the unperturbed matrix estimate, $\eta B_{\mathrm{pert}}(\gamma)$ is the linear loss due to the perturbation, and $C_*(\gamma) \le 1+\sqrt{2}$ is the conversion factor from the modified energy to the Hilbert norm. In unbounded models one must additionally know that the stationary state $h_{\gamma,\eta}$ belongs to the chosen reference space. Corollary 5.2 then optimizes the lower bound over γ under uniformity assumptions on the structural constants.

Corollary 5.2 (Optimal friction and kinetic scaling)**.** *Keep the notation of Theorem 5.1, and assume Assumptions 4.4, 4.5, and 4.6. If one wants to interpret the resulting formulas as centered-semigroup bounds, assume in addition that Assumption 4.1 holds at the frictions under consideration, in particular at γ_{opt}. Assume furthermore that the structural constants Θ, κ_{ell}, k_{reg}, λ_R, $b_{2,\mathrm{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$, that the gap-rescaled lower-order quantity $\kappa_{\mathrm{low}}/\sqrt{\lambda(Q_O)}$ remains bounded as $\lambda(Q_O) \downarrow 0$ (equivalently, $K_{\mathrm{geom}} = \mathcal{O}(1)$; for the Langevin family this is precisely the regime $K \le K_0\lambda(Q_O) = K_0 m$), and that $b_{2,\mathrm{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ defined in (49) attains its global maximum at

$$\gamma_{\mathrm{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \asymp \sqrt{\lambda(Q_O)}. \tag{56}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{57}$$

Moreover, the mixing weight in (50) satisfies

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{58}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\mathrm{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and the certified threshold satisfies

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{59}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{60}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\mathrm{reg}}$. Then $\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\mathrm{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (50), $B_{\mathrm{pert}}(\gamma) = b_{2,\mathrm{eff}} + \alpha_*(\gamma)\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\mathrm{opt}}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\mathrm{opt}}) = x + K_{\mathrm{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{Z(\gamma_{\mathrm{opt}})^2+2k_{\mathrm{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Since $b_{2,\mathrm{eff}} > 0$ by assumption, we obtain

$$\begin{aligned} b_{2,\mathrm{eff}} &\le B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) \\ &= b_{2,\mathrm{eff}} + \alpha_*(\gamma_{\mathrm{opt}})\, C_{\mathrm{sing}}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1), \end{aligned} \tag{61}$$

hence $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) \asymp 1$. Finally, the matrix-controlled threshold satisfies $\Lambda_{\mathrm{hypo}}(\gamma_{\mathrm{opt}})/(2B_{\mathrm{pert}}(\gamma_{\mathrm{opt}})) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 = 0$, then $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = +\infty$ and therefore $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 > 0$, then $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = 2\gamma_{\mathrm{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$. In either case,

$$\eta^*(\gamma_{\mathrm{opt}}) = \min\left\{\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}),\ \frac{\Lambda_{\mathrm{hypo}}(\gamma_{\mathrm{opt}})}{2B_{\mathrm{pert}}(\gamma_{\mathrm{opt}})}\right\} \asymp \sqrt{\lambda(Q_O)}.$$

□

For the equilibrium Langevin warm-up of Section 2, the general formulas reproduce the same $\sqrt{\lambda(Q_O)}$ scaling. Proposition 2.2–Corollary 2.3 retain the combined Langevin constant $\sqrt{2 + \frac{K}{2m}}$, whereas the abstract argument passes through the split quantity $K_{\mathrm{geom}} = \kappa_{\mathrm{ell}} + \kappa_{\mathrm{low}}/(2\sqrt{\lambda(Q_O)})$.

5.2. **Diffusive upper bound in the overdamped regime.** We now turn to the complementary upper bound. A contractive semigroup estimate on $\mathcal{H}_0$ converts a bound $\|g_t\| \le Ce^{-\nu t}\|g\|$ into a bound on the injectivity modulus of $\mathcal{L}^\dagger$ in the same reference norm.

Define the *singular-value gap of the centered forward generator*

$$s(\mathcal{L}^\dagger) := s(\mathcal{L}_0^\dagger) := \inf_{g\in\mathcal{H}_0\cap\mathrm{dom}(\mathcal{L}^\dagger)\setminus\{0\}} \frac{\|\mathcal{L}^\dagger g\|}{\|g\|}, \tag{62}$$

where $\mathcal{L}_0^\dagger$ is the part generator from Lemma 4.3. This is the smallest singular value of the centered generator in finite dimensions and the corresponding injectivity modulus in the unbounded setting.

Assumption 5.3 (Additional hypotheses for the upper bound)**.** In addition to Assumptions 4.1, 4.4, 4.5, and 4.6, we assume:

(i) $\ker(\mathcal{L}^\dagger) \cap \mathcal{H}_0 = \{0\}$;
(ii) $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ (automatic in finite dimensions and many compact-resolvent settings);
(iii) the first-order corrector $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$ belongs to $\mathrm{dom}(\mathcal{R}) \cap \mathrm{dom}((-\mathcal{R})^{1/2}) \cap \mathrm{dom}(\mathcal{V}_{\mathrm{skew}})$, the test vector $g_\gamma := \phi + \gamma^{-1}\psi$ belongs to $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathrm{dom}(\mathcal{V}_{\mathrm{pert}}^\dagger)$ for the parameter pair under consideration, and $P_F\mathcal{V}_{\mathrm{skew}}\psi$ is well defined;

These items provide a slow eigenmode and its first-order fast corrector. The semigroup-to-resolvent step will instead use the assumed exponential decay estimate itself.

Item (iii) is the only genuinely model-specific regularity input in the ceiling theorem. In Example 6.2 it follows from the explicit $\mathscr{S}$-core identities verified in Appendix E, while in Example 6.3 it is automatic because every operator is bounded.

Theorem 5.4 (Diffusive upper bound in the overdamped regime)**.** *Fix the parameter pair (γ, η) and assume Assumption 5.3. Let $C \geq 1$ and $\nu > 0$ be such that*

$$\|e^{t\mathcal{L}^\dagger} g\| \leq C\, e^{-\nu t}\, \|g\|, \qquad \forall g \in \mathcal{H}_0,\ \forall t \geq 0. \tag{63}$$

Assume moreover that the semigroup on $\mathcal{H}_0$ is contractive:

$$\|e^{t\mathcal{L}^\dagger} g\| \leq \|g\|, \qquad \forall t \geq 0,\ \forall g \in \mathcal{H}_0. \tag{64}$$

Then

$$\begin{aligned} \nu &\leq (1 + \log C)\, s(\mathcal{L}^\dagger) \\ &\leq (1 + \log C) \left[\frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O) + \kappa_{\mathrm{low}} \sqrt{\lambda(Q_O)}}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \end{aligned} \tag{65}$$

where κ_{ell} and κ_{low} are the flux-regularity constants from Assumption 4.5.3 and a_1, a_2, b_1, b_2 are from Assumption 4.5.4. Consequently, along any family of parameter pairs satisfying the hypotheses uniformly and obeying $\eta \leq c/\gamma$, the right-hand side has diffusive overdamped scale $\mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$.

By Lemma 4.2, the same ceiling applies to convergence of f_t toward $h_{\gamma,\eta}$ for initial data in $\mathcal{H}_1$.

Proof. Corollary D.2 shows that every contractive exponential estimate on $\mathcal{H}_0$ satisfies

$$\nu \leq (1 + \log C)\, s(\mathcal{L}^\dagger).$$

It therefore remains to bound the singular-value gap $s(\mathcal{L}^\dagger)$. For that step, test $\mathcal{L}^\dagger$ on the first-order slow-mode ansatz built from an eigenfunction ϕ of Q_O and its fast corrector $\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi$. The two-term flux estimate from Assumption 4.5.3 controls the only nontrivial fast remainder, and Assumption 4.5.4 controls the perturbation through the $(-\mathcal{R})^{1/2}$-energy bounds. This yields the displayed upper bound on $s(\mathcal{L}^\dagger)$ and therefore on ν. The detailed singular-value computation is given in Appendix D. □

Thus Theorem 5.4 gives the explicit overdamped $1/\gamma$ upper bound in the same fixed L^2 reference norm, with factor $1 + \log C$ under contractivity.

For families with $\kappa_{\mathrm{low}}/\sqrt{\lambda(Q_O)} = \mathcal{O}(1)$, the $1/\gamma$ coefficient is of order $\lambda(Q_O)$. Hence the upper bound has the same family scaling as the window $\gamma \asymp \sqrt{\lambda(Q_O)}$ from Corollary 5.2.

6. Examples

This section has three roles. Section 6.1 records a short operator-theoretic recipe for constructing centered second-order lifts from a prescribed effective operator. Example 6.2 verifies the abstract hypotheses in the main unbounded setting and states the additional stationary-state input needed there. Example 6.3 is a finite-dimensional toy inspired by boundary-driven/Zeno models. All figures and tables are numerical illustrations only.

6.1. Constructing a lift from a prescribed effective operator.

We record a compact construction. Start from a prescribed effective operator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$ and seek a lifted operator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ whose adiabatic elimination (Definition 3.3) is exactly $\mathcal{L}_O$. This runs in the opposite direction from physical Schrieffer–Wolff/Zeno reductions. Kessler's generalized Schrieffer–Wolff formalism [18] constructs effective dissipative generators perturbatively, while [4, 3] analyze specific boundary-driven Lindblad regimes. Here, by contrast, we start from the target slow operator and engineer a lift with the desired Schur complement. Unless

positivity constraints are imposed separately, the recipe below need not preserve Markov, positivity, complete positivity, GKLS, or Lindblad structure.

The recipe has three steps. Writing $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ in Definition 3.3 gives

$$\begin{aligned}\mathcal{L}_O = &\underbrace{-P_S\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\,P_S}_{Q_O}\\ &\underbrace{-\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)}\\ &\underbrace{-\eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}.\end{aligned} \tag{66}$$

Step 1 extracts the operator Q_O at $\eta = 0$; Step 3 matches the remaining $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ terms.
Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The result is Q_O, whose spectral gap $\lambda(Q_O)$ is the basic scale in Theorem 5.1.
Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = T^\dagger\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\text{skew}}P_S, \tag{67}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^\dagger\mathcal{R}_F^{-1}T = Q_O$. A universal bounded-model choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then $T^\dagger\mathcal{R}_F^{-1}T = Q_O$, and we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\text{skew}} := \left(\begin{smallmatrix} 0 & -T^\dagger \\ T & 0 \end{smallmatrix}\right)$. This is the choice used in the finite-dimensional algorithmic-lift comparisons below. In genuinely unbounded settings it should be regarded only as a formal template unless one separately proves the relevant domain, closedness, and generator properties for the block operator.
Step 3: Determine $\mathcal{V}_{\text{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\text{skew}}$ fixed, we choose $\mathcal{V}_{\text{pert}}$ so that adiabatic elimination reproduces $\mathcal{L}_O$. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O \;=\; -\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S \;-\; \eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S, \tag{68}$$

with the centering constraint $P_S\mathcal{V}_{\text{pert}}P_S = 0$. In the Langevin example, $\mathcal{V}_{\text{pert}}P_S = 0$, so the consistency relation simplifies strongly; in the two-qubit toy all terms are present. In finite dimensions this is an algebraic, generally non-unique problem; in infinite dimensions solvability is model-dependent. The additional choice $P_F\mathcal{V}_{\text{pert}}P_F = 0$ imposed below is optional and is used only to remove projection bias.

Proposition 6.1 (Zero projection bias for designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^\dagger \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (68)*. Let g_η denote the normalized stationary vector of the effective slow forward generator $\mathcal{L}_O^\dagger$, assumed unique. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the adjoint Schur complement

$$S^\dagger(\gamma) := -P_S\mathcal{V}^\dagger\big(\gamma\mathcal{R}_F + P_F\mathcal{V}^\dagger P_F\big)^{-1}\mathcal{V}^\dagger P_S \tag{69}$$

satisfies $\gamma\,S^\dagger(\gamma) = \mathcal{L}_O^\dagger$ for all $\gamma > 0$;
(iii) whenever the lifted stationary vector $h_{\gamma,\eta} \in \mathcal{H}_1 \cap \ker(\mathcal{L}^\dagger)$ and the effective stationary vector $g_\eta \in \mathcal{H}_S \cap \mathcal{H}_1 \cap \ker(\mathcal{L}_O^\dagger)$ are well defined, the lifted stationary vector satisfies $P_S h_{\gamma,\eta} = g_\eta$ for all $\gamma > 0$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F\mathcal{V}_{\text{pert}}P_F = 0$ is a design choice. Indeed, (68) *constrains only the off-diagonal blocks, so the fast–fast block may be set to zero without changing the effective dynamics.*

Proof. (i) By construction, $\mathcal{V}_{\text{skew}}$ is block-off-diagonal. Hence

$$P_F \mathcal{V}_{\text{skew}} P_F = 0.$$

Combined with $P_F \mathcal{V}_{\text{pert}} P_F = 0$, this gives

$$P_F \mathcal{V} P_F = P_F(\mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}) P_F = 0.$$

(ii) With $P_F \mathcal{V} P_F = 0$, one also has $P_F \mathcal{V}^\dagger P_F = 0$, so the adjoint Schur complement reduces to

$$S^\dagger(\gamma) = -P_S \mathcal{V}^\dagger (\gamma \mathcal{R}_F)^{-1} \mathcal{V}^\dagger P_S = \gamma^{-1} \mathcal{L}_O^\dagger.$$

(iii) Let $u := P_S h_{\gamma,\eta}$. Eliminating the fast component of the stationary equation $\mathcal{L}^\dagger h_{\gamma,\eta} = 0$ gives the adjoint Schur-complement equation $S^\dagger(\gamma) u = 0$. Since $\gamma S^\dagger(\gamma) = \mathcal{L}_O^\dagger$, this becomes $\mathcal{L}_O^\dagger u = 0$. By the uniqueness and normalization of g_η, this forces $u = g_\eta$. □

Proposition 6.1 gives the design rule used later: if $P_F \mathcal{V}_{\text{pert}} P_F = 0$, then the projected lifted stationary vector equals the stationary vector of the effective forward generator $\mathcal{L}_O^\dagger$ for all γ. In general the slow marginal $u = P_S h_{\gamma,\eta}$ of a lifted stationary vector satisfies the adjoint Schur complement equation

$$S^\dagger(\gamma)\, u = 0, \qquad S^\dagger(\gamma) := -P_S \mathcal{V}^\dagger \big(\gamma \mathcal{R}_F + P_F \mathcal{V}^\dagger P_F\big)^{-1} \mathcal{V}^\dagger P_S. \tag{70}$$

Under standard analytic perturbation hypotheses for the simple stationary eigenvalue of the adjoint Schur complement, one expects a model-dependent first-order estimate of the form $\|P_S h_{\gamma,\eta} - g_\eta\| = \mathcal{O}(\eta/\gamma)$. When $P_F \mathcal{V} P_F = 0$, as in Proposition 6.1, the bias vanishes identically. This is the construction used in the numerical algorithmic lifts below.

6.2. **Nonequilibrium Langevin family.** This is the main unbounded test case. Proposition 6.3 verifies the abstract hypotheses with constants uniform in m once $K \le K_0 m$. The extra model-specific input is Assumption 6.2, which ensures that the stationary state belongs to the chosen reference space. Combined with $\eta \le \eta^*(\gamma)$, it gives the rigorous parameter regime for Theorem 5.1. The figures later in the subsection come from truncated models and are included only as illustrations.

We consider a nonequilibrium underdamped Langevin family on $\mathbb{R}^{2d}$ with confining potential, exact position spectral gap $m > 0$, lower Hessian bound $\nabla^2 U \ge -K\,\mathrm{Id}$, and a non-gradient forcing satisfying the relative-bounded perturbation hypothesis (78). The quantitative constants in the abstract assumptions are independent of m, while the transport constant after the choice $\varpi_* = \lambda(Q_O) = m$ remains uniform whenever $K \le K_0 m$. In particular, $\kappa_{\text{low}} \le \sqrt{2K}$, so the lower-order defect disappears in the convex case. Throughout this example we fix $d \ge 2$, $\beta > 0$, $m \in (0, 1]$, and $K \ge 0$, and take $U \in C^\infty(\mathbb{R}^d)$ to satisfy

$$\nabla^2 U(q) \ge -K\,\mathrm{Id} \quad \text{for all } q \in \mathbb{R}^d, \tag{71}$$

with all derivatives of positive order growing at most polynomially at infinity. We also take $F \in C^\infty(\mathbb{R}^d; \mathbb{R}^d)$ to be not of gradient form, with all derivatives of positive order growing at most polynomially at infinity. The polynomial-growth assumptions on U and F are used only to place the model within the standard confining Kramers–Fokker–Planck core framework; the explicit constants below enter only through the relative bounds (78). In particular, we do not assume that F is bounded.

Rigorous verification. Setup. Let

$$d\mu(q,p) = Z^{-1} \exp\Big(-\beta\big(\tfrac{1}{2}|p|^2 + U(q)\big)\Big)\, dq\, dp = d\mu_q(q)\, d\mu_p(p), \tag{72}$$

where $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$, and work in $\mathcal{H} = L^2(\mu)$ with slow projection

$$P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\, d\mu_p(p). \tag{73}$$

We assume that the position marginal μ_q has exact spectral gap $m > 0$, equivalently that the optimal Poincaré constant is m^{-1}:

$$\beta^{-1}\|\nabla_q\phi\|^2_{L^2(\mu_q)} \geq m\Big\|\phi - \int \phi\, d\mu_q\Big\|^2_{L^2(\mu_q)}, \qquad \phi \in H^1(\mu_q). \tag{74}$$

Uniform convexity $\nabla^2 U \geq m\,\mathrm{Id}$ is one sufficient special case by Bakry–Émery, but the verification below only uses (74) together with the lower Hessian bound. We use the Schwartz core

$$\mathcal{C}_{\mathrm{Lan}} := \mathscr{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p). \tag{75}$$

The appendix explains the common-core input used for both $\mathcal{L}$ and $\mathcal{L}^\dagger$: the equilibrium part is covered by standard Kramers–Fokker–Planck theory, and the perturbation preserves $\mathcal{C}_{\mathrm{Lan}}$ because F has polynomially growing coefficients. The additional C^∞ and polynomial-growth assumptions on U and F are used only here: they place the model in the standard Kramers–Fokker–Planck core theory, so that the abstract L^2 argument can be justified on $\mathcal{C}_{\mathrm{Lan}}$. At the level of Assumption 4.5, the quantitative constants below depend only on $(\beta, A_\nabla, B_\nabla, A_p, B_p, K)$ and are independent of m; after choosing $\varpi_* = \lambda(Q_O) = m$, the effective transport quantity K_{geom} depends only on K/m. In the regime $K \leq K_0 m$, it is therefore controlled solely by K_0. We apply the abstract theorem on the explicit reference mass-zero space

$$L^2_0(\mu) := \Big\{g \in L^2(\mu) : \int_{\mathbb{R}^{2d}} g\, d\mu = 0\Big\}, \qquad L^2_1(\mu) := \mathbf{1} + L^2_0(\mu).$$

Assumption 6.2 (Additional stationary-state input for the confining Langevin family)**.** There exists $c_{\mathrm{NESS}} > 0$ such that, whenever $|\eta| \leq c_{\mathrm{NESS}} \min(1,\gamma)$, the invariant measure of the nonequilibrium Langevin dynamics can be written as

$$\psi_{\gamma,\eta} = h_{\gamma,\eta}\,\mu, \qquad h_{\gamma,\eta} \in L^2_1(\mu) \cap \mathrm{dom}(\mathcal{L}^\dagger),$$

with

$$\mathcal{L}^\dagger h_{\gamma,\eta} = 0.$$

Moreover the centered correction

$$r_{\gamma,\eta} := h_{\gamma,\eta} - \mathbf{1} \in L^2_0(\mu)$$

depends analytically on η near 0; equivalently, $r_{\gamma,\eta}$ admits a convergent Neumann-series expansion in $L^2_0(\mu)$ whose radius of convergence is bounded below by a constant of order $\min(1,\gamma)$.

Assumption 6.2 is the extra model-specific input used in this example. It is the version of Assumption 4.1 that guarantees centering around $h_{\gamma,\eta}$ is legitimate. The periodic analogue on a torus is proved by Iacobucci–Olla–Stoltz [15, §2.3]; we do not claim that theorem here for the present confining $\mathbb{R}^{2d}$ model. Combined with the structural verification below and with $\eta \leq \eta^*(\gamma)$, it gives the rigorous parameter regime for Theorem 5.1 on $L^2_0(\mu)$.

Generator decomposition. The Langevin generator is written as

$$\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\mathrm{pert}}, \tag{76}$$

with

$$\mathcal{V}_{\mathrm{skew}} := p\cdot\nabla_q - \nabla_q U\cdot\nabla_p, \qquad \mathcal{R} := -p\cdot\nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\mathrm{pert}} f := F(q)\cdot\nabla_p f. \tag{77}$$

Instead of a pointwise boundedness assumption on F, we require only the operator-relative bounds

$$\begin{aligned} \|F(q)\cdot\nabla_p f\|_{L^2(\mu)} &\leq A_\nabla\,\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + B_\nabla\,\|f\|_{L^2(\mu)}, \\ \|F(q)\cdot p\, f\|_{L^2(\mu)} &\leq A_p\,\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)} + B_p\,\|f\|_{L^2(\mu)}, \end{aligned} \tag{78}$$

for every $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$, with constants $A_\nabla, B_\nabla, A_p, B_p \geq 0$ independent of m and K. This is the precise model-specific input corresponding to Assumption 4.5.4. Bounded forces are a sufficient special case, but the verification below uses only (78).

Effective generator $\mathcal{L}_O$ (explicit formula). Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i \, d\mu_p = 0$, Definition 3.3 gives, for $\phi = \phi(q)$,

$$\mathcal{L}_O \phi = \beta^{-1} \Delta_q \phi - (\nabla_q U - \eta F(q)) \cdot \nabla_q \phi, \tag{79}$$

and therefore

$$Q_O \phi = \beta^{-1} \Delta_q \phi - \nabla_q U \cdot \nabla_q \phi. \tag{80}$$

The slow reference measure is $\mu_q(dq) \propto e^{-\beta U(q)} \, dq$. By the assumed Poincaré inequality (74),

$$-\langle \phi, Q_O \phi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \phi\|^2_{L^2(\mu_q)} \geq m \, \|\phi - \textstyle\int \phi \, d\mu_q\|^2_{L^2(\mu_q)}. \tag{81}$$

Hence

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = m. \tag{82}$$

Since F is not a gradient field, the drift $\nabla_q U - \eta F(q)$ is not of gradient form for $\eta \neq 0$, so the effective generator $\mathcal{L}_O$ is genuinely nonequilibrium.

Non-self-adjointness on the slow space. In $L^2(\mu_q)$, integration by parts gives

$$(F \cdot \nabla_q)^\dagger = -F \cdot \nabla_q - (\nabla \cdot F) + \beta \, \nabla_q U \cdot F. \tag{83}$$

Hence $\mathcal{L}_O$ is generically non-self-adjoint as soon as $\eta \neq 0$. For the concrete force used in the numerics below,

$$F(q) = (-\tanh q_2, \tanh q_1), \tag{84}$$

one has $\nabla \cdot F = 0$ but $\nabla_q U \cdot F \not\equiv 0$ for any non-radial U, so $\mathcal{L}_O$ is explicitly non-self-adjoint for every $\eta \neq 0$.

Verification of Assumptions 4.4, 4.5, and 4.6. The structural hypotheses hold with quantitative constants depending on $(\beta, A_\nabla, B_\nabla, A_p, B_p, K)$ at the assumption level, and with a transport quantity at $\varpi_* = \lambda(Q_O)$ depending on K/m. In particular, if $K \leq K_0 m$, then K_{geom} is uniform in m.

Proposition 6.3 (Verification for the lower-Hessian family)**.** *For the family* (77) *with $U \in C^\infty(\mathbb{R}^d)$ satisfying $\nabla^2 U \geq -K \, \mathrm{Id}$ and with all derivatives of positive order growing at most polynomially, whose position marginal μ_q has spectral gap $m > 0$, and force F satisfying* (78)*, Assumptions 4.4, 4.5, and 4.6 hold in $L^2(\mu)$. Moreover, one may choose structural constants satisfying*

$$\lambda_R = 1, \qquad M_R = 1, \qquad \kappa_{\mathrm{ell}} \leq \sqrt{2}, \qquad \kappa_{\mathrm{low}} \leq \sqrt{2K}, \tag{85}$$

and one may take

$$a_1 = A_\nabla + \tfrac{\beta}{2} A_p, \qquad b_1 = B_\nabla + \tfrac{\beta}{2} B_p, \qquad a_2 = \tfrac{\beta}{2} A_p, \qquad b_2 = \tfrac{\beta}{2} B_p. \tag{86}$$

In particular, the perturbation constants (a_1, b_1, a_2, b_2) depend only on $(\beta, A_\nabla, B_\nabla, A_p, B_p)$ and are independent of (m, K). More precisely, the appendix proves the weighted elliptic estimate in the form

$$\|P_F \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) w_\psi\|^2 \leq 2 \, \|(-Q_O)\psi\|^2 + 2K \, \|(-Q_O)^{1/2} \psi\|^2, \tag{87}$$

hence

$$\|P_F \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) w_\psi\| \leq \sqrt{2} \, \|(-Q_O)\psi\| + \sqrt{2K} \, \|(-Q_O)^{1/2} \psi\|, \tag{88}$$

for slow inputs ψ. This is exactly Assumption 4.5.3 with the displayed values of κ_{ell} and κ_{low}. In particular, if $K \le K_0 m$, then the transport factor at $\varpi_* = \lambda(Q_O)$ satisfies

$$K_{\mathrm{geom}} = k_{\mathrm{dual}}(\lambda(Q_O)) \le \sqrt{2} + \sqrt{K_0/2} \tag{89}$$

uniformly in m; the special case $K = m$ gives $K_{\mathrm{geom}} \le \sqrt{2} + 1/\sqrt{2}$. This abstract constant is expressed in the split two-term form of the general theorem rather than the recombined warm-up factor $\sqrt{2 + \frac{K}{2m}}$ from Proposition 2.2. The extra polynomial-growth assumptions on the derivatives of U enter only in the qualitative Schwartz-core verification recalled in Appendix E; they do not affect these quantitative constants.

Corollary 6.4 (Structural kinetic scaling for the lower-Hessian family)**.** *Fix $K_0 \ge 0$ and assume $K \le K_0 m$. For the Langevin family above, the hypotheses controlling γ_{opt} and $\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma)$ in Corollary 5.2 are satisfied uniformly as $m \downarrow 0$. Consequently,*

$$\gamma_{\mathrm{opt}} \asymp \sqrt{m}, \qquad \max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{m}. \tag{90}$$

If, in addition, Assumption 6.2 holds on the parameter range under consideration and the family satisfies the extra threshold hypothesis $b_{2,\mathrm{eff}} > 0$, then the DMS-certified forcing window also obeys

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{m}. \tag{91}$$

Thus Example 6.2 gives a structural verification of the $\sqrt{\lambda(Q_O)}$ acceleration mechanism, including the concrete lower-Hessian regime $\nabla^2 U \ge -m\,\mathrm{Id}$, and yields the same semigroup scaling once the additional stationary-state input is supplied.

The proof of Proposition 6.3 is given in Appendix E. Together with Assumption 6.2, it yields the assumptions of Theorem 5.1 for this family on the intersection of the realization and DMS windows, and it gives the kinetic $\sqrt{m}$ scaling described above whenever $K \le K_0 m$.

Truncated numerical illustrations. To illustrate the theorem, we test the quadratic instance $U(q) = \frac{m}{2}|q|^2$ in $d = 2$, with $F(q) = (-\tanh q_2, \tanh q_1)$ and $\eta = c\sqrt{m}$, $c \in \{0, \frac{1}{2}, 1\}$. The generator is discretized in a tensor-product Hermite–Galerkin basis of dimension 256. These computations are illustrative only.

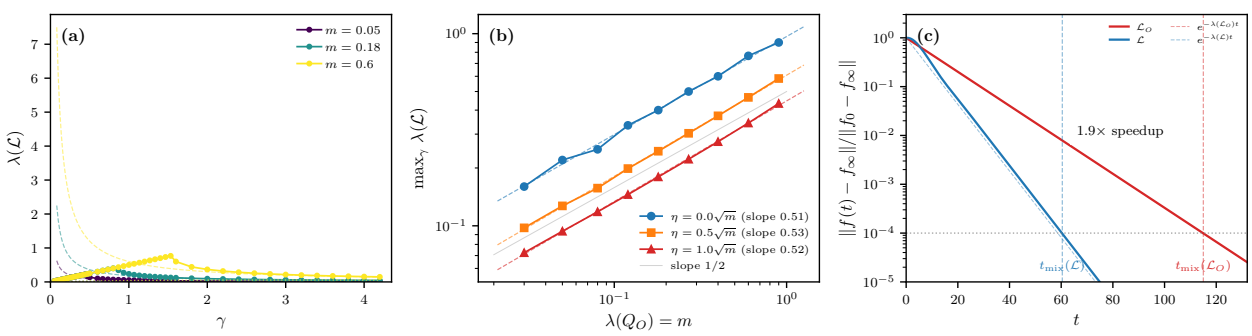

FIGURE 1. Numerical experiment on the truncated Hermite–Galerkin model for Example 6.2 ($\beta = 1$, $m \in [0.03, 0.90]$, 45-point γ-grid). **(a)** Spectral gap versus γ at $\eta = 0$; dashed: overdamped envelope m/γ; dotted: DMS lower bound with $\varpi_* = m$. **(b)** Peak gap versus m on log–log scale for $c \in \{0, \frac{1}{2}, 1\}$; fitted slopes ≈ 0.51–0.53. **(c)** Semigroup decay at $m = 0.08$, $c = \frac{1}{2}$, $\gamma^* = 0.5$; vertical markers: ε-mixing times ($\varepsilon = 10^{-4}$).

Figure 1 shows the three-regime structure and the predicted $\sqrt{\lambda(Q_O)}$ scaling: on the sampled γ-grid, the fitted log–log slopes of the peak gaps are 0.51, 0.53, and 0.52 for $c = 0$, $\frac{1}{2}$, 1. At $m = 0.08$,

$c = \frac{1}{2}$, the truncated lifted gap $\lambda(\mathcal{L}) \approx 0.157$ gives about a 1.9× speedup over $\lambda(\mathcal{L}_O) \approx 0.080$, with mixing times $t_{\text{mix}} \approx 60$ versus 115.

Algorithmic lift via generic square-root factorization. We next apply the universal construction from Section 6.1 to the truncated slow Hermite space. This yields a 32×32 algorithmic lift instead of the 256×256 physical lift. Since $\mathcal{V}_{\text{pert}} P_S = 0$ and $P_F \mathcal{V}_{\text{pert}} P_F = 0$, the consistency relation simplifies and Proposition 6.1 applies. Figure 2 and Table 1 show that the algorithmic and physical lifts track each other closely: the fitted log–log slopes agree, and the observed mixing times are essentially the same.

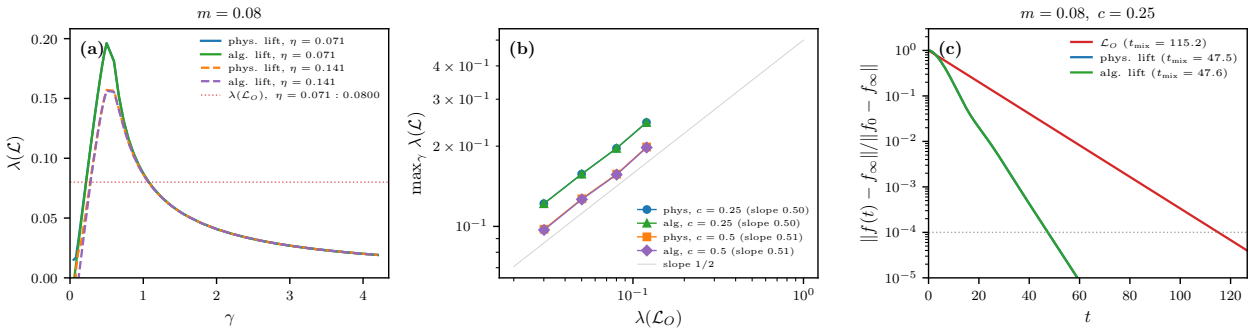

FIGURE 2. Comparison of the *physical* (256×256) and *algorithmic* (32×32, square-root factorization) lifted generators for the truncated model from Example 6.2. **(a)** Spectral gap versus γ at $m = 0.08$ for both lifts with $c = \frac{1}{4}$ and $c = \frac{1}{2}$ (that is, $\eta = c\sqrt{m}$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap versus $\lambda(\mathcal{L}_O)$ on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$ for both lifts. **(c)** Convergence comparison at $m = 0.08$, $c = \frac{1}{4}$, $\varepsilon = 10^{-4}$: algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 1. Numerically computed peak spectral gaps obtained by maximizing over the sampled γ-grid for the physical and algorithmic lifts on the matched grid. The final column reports the ratio of the sampled physical peak to $\lambda(\mathcal{L}_O)$.

m	c	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.03	$\frac{1}{4}$	0.03001	0.12175	0.12163	4.06×
0.03	$\frac{1}{2}$	0.03010	0.09757	0.09701	3.24×
0.05	$\frac{1}{4}$	0.05001	0.15718	0.15703	3.14×
0.05	$\frac{1}{2}$	0.05016	0.12696	0.12629	2.53×
0.08	$\frac{1}{4}$	0.08002	0.19628	0.19613	2.45×
0.08	$\frac{1}{2}$	0.08024	0.15707	0.15640	1.96×
0.12	$\frac{1}{4}$	0.12002	0.24546	0.24519	2.05×
0.12	$\frac{1}{2}$	0.12032	0.19855	0.19765	1.65×

6.3. **Two-qubit example.** We keep the second example as a finite-dimensional toy. It is inspired by boundary-driven/Zeno models, but its role here is only to make the abstract assumptions completely explicit. Example 6.2 remains the main confining PDE test case.

All operators are bounded, the structural hypotheses reduce to linear algebra, and the reduced slow dynamics is already non-self-adjoint in the reference inner product for $\eta \neq 0$. In finite dimensions, once $\ker(Q_O) = \mathbb{C}\mathbf{1}$, the fast-to-slow map factors directly through $-Q_O$ on the mean-zero

slow space, so the appendix proves the stronger one-term estimate

$$\kappa_{\text{low}} = 0. \tag{92}$$

Thus the revised two-term regularity assumption does not enlarge the proof burden in this example.

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$. Define

$$L_0^2(\sigma) := \{g \in L^2(\sigma) : \langle g, \mathbf{1}\rangle_\sigma = 0\}, \qquad L_1^2(\sigma) := \mathbf{1} + L_0^2(\sigma).$$

Here $\mathcal{L}\mathbf{1} = 0$, so $L_0^2(\sigma)$ is invariant under $e^{t\mathcal{L}^\dagger}$. The structural verification below concerns Assumptions 4.4, 4.5, and 4.6. This example is only boundary-driven/Zeno-inspired: for $\eta \neq 0$, the perturbation $\sigma_z \otimes \mathcal{D}^\dagger[\sigma_-]$ is not $*$-preserving, so no positivity, complete positivity, GKLS, or Lindblad claim is made for the full toy family or for the reduced slow generator. Whenever a normalized stationary vector $h_{\gamma,\eta} \in L_1^2(\sigma) \cap \ker(\mathcal{L}^\dagger)$ is fixed for the parameter pair under consideration, the centered theorem is then applied on $L_0^2(\sigma)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, where $\Gamma > 0$, so $\lambda_R = 4\Gamma$. The transport part uses a *two-channel* coupling,

$$\mathcal{V}_{\text{skew}} X = -i[g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z), X], \tag{93}$$

and the perturbation is $\mathcal{V}_{\text{pert}} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. The fast block $\mathcal{R}$ is a depolarizing Lindbladian on the ancilla, but away from $\eta = 0$ the coupled family is used here only as a bounded operator toy. Because $\mathrm{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$. Because all operators are bounded in this 16×16 model, the three-step computation below is an exact evaluation of Definition 3.3, not a formal heuristic.

Step 1 (first excursion to fast space). The two-channel coupling gives

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) + \sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{94}$$

Step 2 (Poisson equation on the fast ancilla). Since $\mathcal{R}_F(\sigma_a) = -\kappa\,\sigma_a$ with $\kappa := 4\Gamma$ for each $a \in \{x, y, z\}$, we have $\mathcal{R}^+(\sigma_a \otimes Y) = -(1/\kappa)\sigma_a \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa}\sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) - \frac{1}{\kappa}\sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{95}$$

Step 3 (return map and projection). For $Y \in \mathcal{B}(\mathfrak{h}_S)$ one has the exact identities

$$\begin{aligned} \mathcal{V}_{\text{skew}}(\sigma_x \otimes Y) &= -ig\,\mathbf{1}_F \otimes [\sigma_x, Y] + g\,\sigma_y \otimes \{\sigma_z, Y\}, \\ \mathcal{V}_{\text{skew}}(\sigma_z \otimes Y) &= -ig\,\mathbf{1}_F \otimes [\sigma_z, Y] - g\,\sigma_y \otimes \{\sigma_x, Y\}. \end{aligned} \tag{96}$$

Thus the cross-channel contributions carry a fast σ_y factor and vanish after applying P_S, since $\mathrm{tr}(\sigma_y) = 0$. Only the same-channel commutator terms survive in the slow projection. Projecting gives

$$\begin{aligned} \mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(& g^2 \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]] \\ & + i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \\ & + i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \\ & - \eta^2\,\mathcal{G}_S^2(X_S)\Big). \end{aligned} \tag{97}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]]. \tag{98}$$

Non-self-adjointness on the slow space. Since the slow marginal of σ is proportional to the identity, the $L^2(\sigma)$ inner product on $\mathcal{H}_S$ reduces to the Hilbert–Schmidt inner product, and $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}[\sigma_-] \neq \mathcal{D}^\dagger[\sigma_-]$. Hence the $\mathcal{O}(\eta)$ correction in $\mathcal{L}_O - Q_O$ is already non-self-adjoint for every $\eta \neq 0$.

This can be seen explicitly on the Pauli basis. Using

$$\mathcal{G}_S(\mathbf{1}) = 0, \qquad \mathcal{G}_S(\sigma_x) = -\tfrac{1}{2}\sigma_x, \qquad \mathcal{G}_S(\sigma_y) = -\tfrac{1}{2}\sigma_y, \qquad \mathcal{G}_S(\sigma_z) = -(\mathbf{1} + \sigma_z), \tag{99}$$

the effective generator (97) acts as

$$\begin{aligned} \mathcal{L}_O(\mathbf{1}) &= 0, \\ \mathcal{L}_O(\sigma_x) &= -\frac{4g^2 - \eta^2/4}{\kappa}\,\sigma_x - \frac{2\eta g}{\kappa}\,\sigma_y, \\ \mathcal{L}_O(\sigma_y) &= \frac{2\eta g}{\kappa}\,\sigma_x - \frac{8g^2 - \eta^2/4}{\kappa}\,\sigma_y, \\ \mathcal{L}_O(\sigma_z) &= \frac{\eta^2}{\kappa}\,\mathbf{1} - \frac{4g^2 - \eta^2}{\kappa}\,\sigma_z. \end{aligned} \tag{100}$$

The σ_x–σ_y coupling and the $\sigma_z \to \mathbf{1}$ drift vanish when $\eta = 0$; for $\eta \neq 0$ they provide an explicit algebraic witness that the reduced dynamics is already non-self-adjoint in the $L^2(\sigma)$ inner product.

Verification of Assumptions 4.4, 4.5, and 4.6. All operators are bounded on the finite-dimensional space $\mathcal{H}$, so the verification reduces to linear algebra.

Proposition 6.5 (Verification for the two-qubit operator toy)**.** *For the two-qubit operator toy above with $\Gamma > 0$, $g \neq 0$, and the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$, Assumptions 4.4, 4.5, and 4.6 hold. Moreover,*

$$\lambda_R = 4\Gamma, \qquad M_R = \|\mathcal{R}\|, \qquad \kappa_{\text{low}} = 0, \qquad \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = g^2/\Gamma, \tag{101}$$

while κ_{ell} is finite and depends only on the finite-dimensional channel geometry. The perturbation constants (a_1, b_1, a_2, b_2) satisfy $a_1 = a_2 = 0$, $b_j = \|\mathcal{K}_{\text{pert}}\|$ or $\|\mathcal{S}_{\text{pert}}\|$.

The proof is given in Appendix F. Thus Example 6.3 verifies the abstract hypotheses on $L^2_0(\sigma)$. Whenever Assumption 4.1 is verified for a chosen parameter pair, Theorem 5.1 applies on $L^2_0(\sigma)$, and Assumption 5.3(iii) is automatic because every operator is bounded in this 16×16 setting. As in Example 6.2, the final convergence statement still depends on the stationary-state input for the chosen parameter pair. The remaining figures and tables are numerical illustrations.

Table 2. Kernel compatibility and spectral gap for Q_O with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$. For all $g > 0$: $\dim\ker(Q_O) = 1$ $(= \mathbb{C}\mathbf{1})$ and $\lambda(Q_O) = g^2/\Gamma$.

g	$\dim\ker(Q_O)$	$\lambda(Q_O)$
0.3	1	0.1125
0.5	1	0.3125
1.0	1	1.2500

Finite-dimensional numerical illustrations. We assemble the exact 16×16 operator family and scan spectral gaps over $g \in \{0.1, \ldots, 1.0\}$, $\Gamma = 0.8$, $\gamma \in [0.02, 7.0]$, and $\eta \in \{0, 0.10, 0.15\}$.

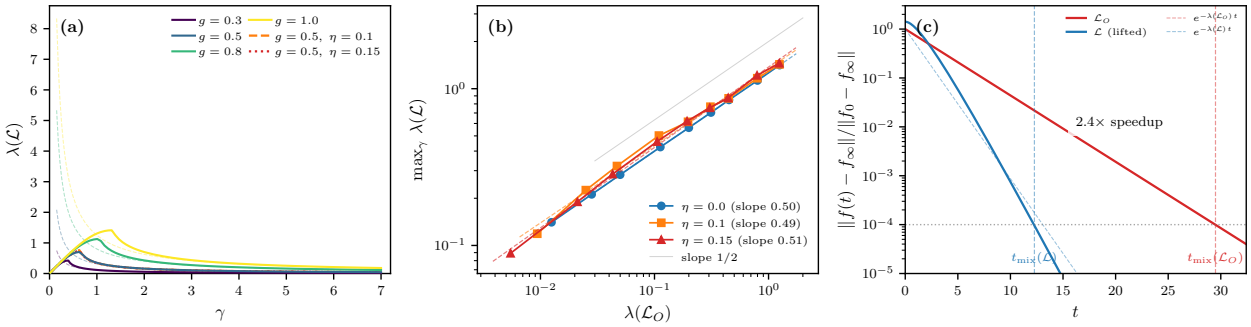

FIGURE 3. Numerics for Example 6.3. **(a)** Sampled spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of the sampled peak spectral gap versus $\lambda(\mathcal{L}_O)$. On the displayed grid, the fitted slopes stay close to the half-law. **(c)** Decay-time comparison for the projected slow component of the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes. Vertical markers indicate the ε-threshold crossing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.1, \ldots, 1.0\}$, 53-point γ-grid over $[0.02, 7.0]$.

Figure 3 again displays the three-regime structure and the $\sqrt{\lambda(Q_O)}$ law: on the sampled γ-grid, the fitted log–log slopes of the peak gaps are 0.50, 0.49, and 0.51, and at $g = 0.5$, $\eta = 0$, the sampled peak-gap ratio relative to $\lambda(\mathcal{L}_O)$ is about 2.4.

Algorithmic lift via generic square-root factorization. We use the same universal square-root factorization in the two-qubit toy, and for a fair comparison we set the algorithmic fast rate equal to the explicit ancilla rate $\lambda_R = 4\Gamma$. Solving the consistency relation (68) numerically on the finite slow space gives an 8×8 algorithmic lift (versus 16×16 for the explicit two-qubit lift), with final residual below 10^{-13}. Figure 4 and Table 3 show that both lifts remain consistent with square-root scaling on the sampled comparison grid, that the algorithmic lift is visibly closer to the half-law there, and that it produces systematically larger sampled peak gaps in this toy model.

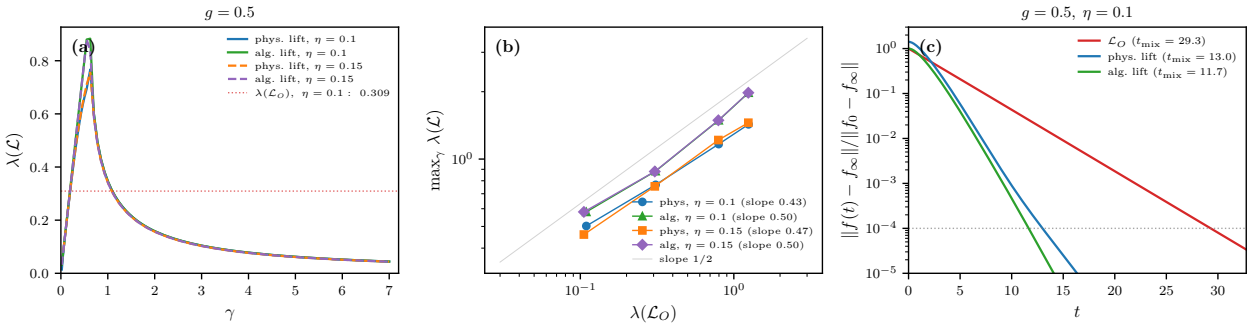

FIGURE 4. Comparison of the *explicit two-qubit* (16×16) and *algorithmic* (8×8, square-root factorization) lifted operators for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of the sampled peak spectral gap versus $\lambda(\mathcal{L}_O)$ for both lifts. **(c)** Decay-time comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, explicit two-qubit lift, and overdamped $\mathcal{L}_O$.

For the *algorithmic* lifts in both examples, $P_F \mathcal{V}_{\text{pert}} P_F = 0$ by construction, so Proposition 6.1 predicts zero projection bias. At the representative comparison points used in the convergence

TABLE 3. Numerically computed peak spectral gaps obtained by maximizing over the sampled 53-point γ-grid for the explicit two-qubit and algorithmic lifts. The final column is the ratio of the sampled explicit-lift peak to $\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	full peak	alg. peak	full ratio
0.3	0.10	0.1094	0.5024	0.5806	4.59×
0.3	0.15	0.1055	0.4600	0.5806	4.36×
0.5	0.10	0.3094	0.7673	0.8831	2.48×
0.5	0.15	0.3055	0.7552	0.8800	2.47×
0.8	0.10	0.7969	1.1663	1.4906	1.46×
0.8	0.15	0.7930	1.2163	1.4906	1.53×
1.0	0.10	1.2469	1.4291	1.9789	1.15×
1.0	0.15	1.2430	1.4534	1.9789	1.17×

panels, $\|P_S h^{\mathrm{alg}}_{\gamma,\eta} - g_\eta\|/\|g_\eta\|$ is at machine precision; the full toy lifts show the same behaviour on the displayed finite-dimensional grids.

7. Conclusion

The central point of the paper is that the factor $\lambda/(\varpi+\lambda)$ in the Dolbeault–Mouhot–Schmeiser corrector determines whether one sees kinetic scaling. Choosing $\varpi = \lambda(Q_O)$ preserves the slowest mode and leads to an explicit lower bound at fixed friction. A complementary upper bound shows that, in the same reference norm, the overdamped regime is diffusive. The Langevin example verifies the abstract hypotheses in the main unbounded setting, while the two-qubit example keeps the mechanism completely explicit in finite dimensions.

Two directions seem especially natural. First, one would like to sharpen the admissible forcing threshold and extend the lower bound beyond the present perturbative regime in η. Second, the lift-construction recipe raises an optimization problem of its own: how should one choose the auxiliary fast space and factorization so that the structural constants remain informative in higher-dimensional or many-body models?

Appendix A. Analytic toolbox: pseudoinverse and Friedrichs construction

This appendix collects the standard functional-analytic constructions underlying the slow–fast framework. In finite dimensions every step is trivial; the statements below are needed only for the unbounded (PDE/diffusion) setting.

Lemma A.1 (Pseudoinverse via spectral calculus)**.** *Let $\mathcal{R}$ be a non-positive self-adjoint operator on $\mathcal{H}$ with $\ker(\mathcal{R}) = \mathcal{H}_S$ and spectral gap $\lambda_R > 0$ on $\mathcal{H}_F = \mathcal{H}_S^\perp$. Let $\{E_\lambda\}_{\lambda\ge 0}$ be the spectral family of $-\mathcal{R}|_{\mathcal{H}_F}$, so that $-\mathcal{R}|_{\mathcal{H}_F} = \int_{\lambda_R}^\infty \lambda\, dE_\lambda$. Then the pseudoinverse $\mathcal{R}^+ := -(-\mathcal{R}|_{\mathcal{H}_F})^{-1}P_F$ is a bounded self-adjoint operator on $\mathcal{H}$ satisfying:*

(i) $\mathcal{R}^+$ maps $\mathcal{H}_F$ into $\mathrm{dom}(\mathcal{R}) \cap \mathcal{H}_F$ and annihilates $\mathcal{H}_S$;
(ii) $\mathcal{R}\mathcal{R}^+ = P_F$ and $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$;
(iii) $\|\mathcal{R}^+\| \le \lambda_R^{-1}$ and $-\mathcal{R}^+ \ge 0$ on $\mathcal{H}_F$;
(iv) the square root satisfies $\|(-\mathcal{R}^+)^{1/2}\| \le \lambda_R^{-1/2}$.

Proof. Since the spectrum of $-\mathcal{R}|_{\mathcal{H}_F}$ is contained in $[\lambda_R, \infty)$, the inverse $(-\mathcal{R}|_{\mathcal{H}_F})^{-1} = \int_{\lambda_R}^\infty \lambda^{-1}\, dE_\lambda$ is a bounded non-negative self-adjoint operator on $\mathcal{H}_F$ with norm at most λ_R^{-1}. The remaining properties follow from the spectral theorem and the identities $E_\lambda \cdot \mathrm{dom}(\mathcal{R}) = \mathrm{dom}(\mathcal{R}) \cap E_\lambda\mathcal{H}$ for each spectral projection. □

Lemma A.2 (Friedrichs construction of the slow operator Q_O)**.** *Under the standing hypotheses on $\mathcal{R}$ and $\mathcal{V}_{\rm skew}$ (including the centered coupling condition $P_S\mathcal{V}_{\rm skew}P_S = 0$), and given a dense subspace $\mathcal{C} \subset \mathrm{dom}(\mathcal{V}_{\rm skew}) \cap \mathcal{H}_S$ that is invariant under $\mathcal{R}^+$ and $P_S\mathcal{V}_{\rm skew}$, define the quadratic form*

$$q_O(\psi) := \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\psi\rangle, \qquad \psi \in \mathcal{C}. \tag{102}$$

Then:

(i) q_O is a densely defined non-negative symmetric form on $\mathcal{H}_S$. Assume moreover that it is closable, and denote its closure again by q_O.

(ii) The Friedrichs extension of the closed form q_O is a non-positive self-adjoint operator Q_O on $\mathcal{H}_S$ satisfying

$$\langle \psi, (-Q_O)\psi\rangle = q_O(\psi), \qquad \forall \psi \in \mathrm{dom}((-Q_O)^{1/2}). \tag{103}$$

(iii) On $\mathcal{C}$, the Friedrichs realization satisfies the weak-form identity

$$\langle \psi, (-Q_O)\psi\rangle = \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\psi\rangle,$$

which is the quadratic-form version of the formal operator expression (24). Any stronger core identity is an additional represented-realization input, imposed later in Assumption 4.4.

(iv) For every $\varpi > 0$, the resolvent $(\varpi - Q_O)^{-1}$ is a bounded self-adjoint operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$. If $\lambda(Q_O) > 0$ (Assumption 4.6), then additionally $\|(\varpi - Q_O)^{-1}\| \le (\varpi + \lambda(Q_O))^{-1}$ on $\mathcal{H}_{S,0}$.

Proof. Define the sesquilinear form

$$\mathfrak{q}_O(\psi, \varphi) := \langle \mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\rm skew}\varphi\rangle, \qquad \psi, \varphi \in \mathcal{C}.$$

Because $P_S\mathcal{V}_{\rm skew}P_S = 0$, one has $\mathcal{V}_{\rm skew}\psi \in \mathcal{H}_F$ for every $\psi \in \mathcal{C} \subset \mathcal{H}_S$, so the form is well defined. Since $-\mathcal{R}^+$ is self-adjoint and non-negative on $\mathcal{H}_F$,

$$\mathfrak{q}_O(\psi, \varphi) = \big\langle (-\mathcal{R}^+)^{1/2}\mathcal{V}_{\rm skew}\psi,\, (-\mathcal{R}^+)^{1/2}\mathcal{V}_{\rm skew}\varphi\big\rangle,$$

hence $\mathfrak{q}_O$ is symmetric and non-negative, and $q_O(\psi) = \mathfrak{q}_O(\psi, \psi)$. The additional closability hypothesis gives a closed non-negative form after closure. Item (ii) is then the standard Friedrichs representation theorem for closed semibounded forms [17], and item (iii) is the corresponding quadratic-form identity on the core. The resolvent bound in item (iv) follows from $\sigma(-Q_O) \subset [0, \infty)$ on $\mathcal{H}_S$, while the sharper bound on $\mathcal{H}_{S,0}$ uses $\sigma(-Q_O|_{\mathcal{H}_{S,0}}) \subset [\lambda(Q_O), \infty)$. □

Appendix B. Proof of Lemma 4.9

This appendix establishes the structural bounds of Lemma 4.9. The proof is organized exactly as the later Lyapunov argument uses it: first the solver and regularity bounds for $\mathcal{A}_\varpi$, then the interaction estimates with $\mathcal{R}$, $\mathcal{V}_{\rm skew}$, and $\mathcal{V}_{\rm pert}$.

Proof. In the unbounded setting, all identities below are first established on the dense invariant core from Section 3.1; the resulting operator bounds then extend to the stated energy spaces by density and continuity. On the core from Assumption 4.4(i), the effective operator on the slow space is represented by

$$Q_O = -P_S\mathcal{V}_{\rm skew}\mathcal{R}^+\mathcal{V}_{\rm skew}P_S.$$

By Assumption 4.5.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{104}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (29), which applies to fluxes of the form $w = w_\psi = P_F\mathcal{V}_{\rm skew}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let f belong to the common core and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\rm skew}u, (-\mathcal{R}^+)P_F f\rangle. \tag{105}$$

Let $v := \mathcal{V}_{\rm skew}u$. Since $P_S\mathcal{V}_{\rm skew}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using Assumption 4.4(ii) with $w_u = P_F\mathcal{V}_{\rm skew}P_S u = v$ and the pseudoinverse bound (104):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{106}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{107}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{108}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{109}$$

The first inequality is the spectral bound $\varpi^{-1/2}\cdot\frac{1}{2}$, which can equivalently be derived from the spectral calculus: writing $Q_O = -\int_0^\infty \lambda\, dE_\lambda$ on $\mathcal{H}_{S,0}$, one has $\|(\varpi - Q_O)^{-1}(-Q_O)^{1/2}\| = \sup_{\lambda\ge 0}\frac{\sqrt{\lambda}}{\varpi+\lambda}$, and the calculus identity $\sup_{\lambda\ge 0}\sqrt{\lambda}/(\varpi+\lambda) = 1/(2\sqrt{\varpi})$ (attained at $\lambda = \varpi$) gives the stated bound. The second inequality uses $\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \lambda_R^{-1/2}\|f\|$. Moreover, $a \le b$. By construction of the resolvent, $u \in \mathrm{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\rm skew}u$ is a slow-induced flux of the form w_ψ with $\psi \in \mathrm{dom}(Q_O)$; we may therefore apply the flux conditioning bound (29) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{110}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. These bounds depend only on $\|f\|$, so the core operator extends uniquely by continuity to a bounded operator $\mathcal{H} \to \mathcal{H}_S$, still denoted $\mathcal{A}_\varpi$. This proves $\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}}\varpi^{-1/2}$ and $\|\mathcal{V}_{\rm skew}\mathcal{A}_\varpi\| \le k_{\rm reg}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\rm reg} := \sqrt{M_R/\lambda_R}. \tag{111}$$

Dual regularity. On the common core, define $\mathcal{T} := \mathcal{A}_\varpi\mathcal{V}_{\rm skew}P_F$. To prove boundedness, it suffices to identify and estimate the operator appearing in the dual pairing against slow test functions. We first compute the adjoint of the bounded solver:

$$\mathcal{A}_\varpi^\dagger = \left((\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\rm skew}P_S(\varpi - Q_O)^{-1}. \tag{112}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$). Consequently, for every core vector f and every $g \in \mathcal{H}_S$,

$$\langle \mathcal{T}f, g\rangle = \left\langle f,\, P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)\mathcal{V}_{\rm skew}P_S(\varpi - Q_O)^{-1}g\right\rangle. \tag{113}$$

We therefore estimate the operator

$$\mathcal{B} := P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)\mathcal{V}_{\rm skew}P_S(\varpi - Q_O)^{-1} : \mathcal{H}_S \to \mathcal{H}_F.$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1}g$. Define the macroscopic flux

$$w_\psi := P_F\mathcal{V}_{\rm skew}P_S\psi \in \mathcal{H}_F. \tag{114}$$

Then $\mathcal{B}g = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.5.3:

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi\| + \kappa_{\text{low}}\,\|(-Q_O)^{1/2}\psi\|. \tag{115}$$

Substituting ψ back:

$$\begin{aligned}\|\mathcal{B}g\| \le\ & \kappa_{\text{ell}}\|(-Q_O)(\varpi - Q_O)^{-1}g\| \\ & + \kappa_{\text{low}}\,\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}g\|.\end{aligned} \tag{116}$$

Since Q_O is self-adjoint and non-positive, functional calculus gives

$$\|(-Q_O)(\varpi - Q_O)^{-1}\| = \sup_{\lambda\ge 0}\frac{\lambda}{\varpi+\lambda} \le 1, \tag{117}$$

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{\lambda\ge 0}\frac{\sqrt{\lambda}}{\varpi+\lambda} = \frac{1}{2\sqrt{\varpi}}. \tag{118}$$

Thus:

$$\|\mathcal{B}g\| \le \left(\kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}}\right)\|g\|. \tag{119}$$

Hence $\mathcal{B}$ extends boundedly from $\mathcal{H}_S$ to $\mathcal{H}_F$ with norm at most $k_{\text{dual}}(\varpi)$. Returning to the dual pairing, we obtain

$$|\langle \mathcal{T}f, g\rangle| \le \|f\|\,\|\mathcal{B}g\| \le k_{\text{dual}}(\varpi)\,\|f\|\,\|g\|, \qquad g \in \mathcal{H}_S,$$

so $\mathcal{T}$ extends uniquely to a bounded operator $\mathcal{H} \to \mathcal{H}_S$ and $\|\mathcal{A}_\varpi\mathcal{V}_{\text{skew}}P_F\| \le k_{\text{dual}}(\varpi)$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$$

from Assumption 4.5, item 4. We also recall the centered mapping property (42), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 4.5, item 4(a)):

$$\|\mathcal{K}_{\text{pert}}f\| \le a_1\,\|(-\mathcal{R})^{1/2}f\| + b_1\,\|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{120}$$

For the symmetric part, we use the energy bound (Assumption 4.5.4(b)):

$$\|\mathcal{S}_{\text{pert}}f\| \le a_2\,\|(-\mathcal{R})^{1/2}f\| + b_2\,\|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{121}$$

Reverse estimates ($\mathcal{V}_{\text{pert}}\mathcal{A}_\varpi$ and $\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}}u\| + \|\mathcal{S}_{\text{pert}}u\| \tag{122}$$

$$\le b_1\,\|u\| + b_2\,\|u\| + (a_1+a_2)\,\|(-\mathcal{R})^{1/2}u\|. \tag{123}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2}u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1+b_2)\,\|u\|. \tag{124}$$

The same estimate holds for $\mathcal{V}_{\text{pert}}u = (\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}})u$, since $u \in \mathcal{H}_S$ implies $(-\mathcal{R})^{1/2}u = 0$ as above. Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}}\varpi^{-1/2}\|f\|$):

$$\|\mathcal{V}_{\text{pert}}\mathcal{A}_\varpi f\|,\ \|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\varpi f\| \le (b_1+b_2)k_{\mathcal{A}}\varpi^{-1/2}\,\|f\|. \tag{125}$$

This establishes (39a) with $k_P(\varpi) := (b_1+b_2)k_{\mathcal{A}}\varpi^{-1/2}$.

Forward estimates ($\mathcal{A}_\varpi \mathcal{V}_{\rm pert}$ and $\mathcal{A}_\varpi \mathcal{V}_{\rm pert}^\dagger$). Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\rm pert}$ and the energy bound on $\mathcal{S}_{\rm pert}$, the same argument applies to $\mathcal{V}_{\rm pert}^\dagger = -\mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$ and to $\mathcal{V}_{\rm pert} = \mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$:

$$\|\mathcal{V}_{\rm pert} w\|,\ \|\mathcal{V}_{\rm pert}^\dagger w\| \le (b_1 + b_2)\,\|w\| + (a_1 + a_2)\,\|(-\mathcal{R})^{1/2} w\|. \tag{126}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\|\,\|g\| \le k_{\mathcal{A}}\,\varpi^{-1/2}. \tag{127}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\rm skew} P_S (\varpi - Q_O)^{-1} g\|. \tag{128}$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$w_\psi := P_F \mathcal{V}_{\rm skew} P_S \psi, \tag{129}$$

$$\|(-\mathcal{R})^{-1/2} w_\psi\|^2 = \langle w_\psi, -\mathcal{R}^+ w_\psi\rangle = -\langle \psi, Q_O \psi\rangle = \|(-Q_O)^{1/2}\psi\|^2, \tag{130}$$

hence

$$\|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} g\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|g\|. \tag{131}$$

(The last step follows from the scalar bound $\sup_{x\ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (126) yields

$$\|\mathcal{V}_{\rm pert}\mathcal{A}_\varpi^\dagger g\|,\ \|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi^\dagger g\| \le \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2}\Big)\,\varpi^{-1/2}\,\|g\|. \tag{132}$$

Since g is arbitrary, this establishes (39b) with $\tilde{k}_P(\varpi) := \big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi^{-1/2}$.

□

Appendix C. Proof of Theorem 5.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 - \alpha \Re\langle \mathcal{A}_\varpi f, f\rangle, \tag{133}$$

where $\mathcal{A}_\varpi$ denotes the bounded extension of the core DMS corrector $(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+) P_F$ associated with the reference effective dynamics Q_O. The proof has three stages: differentiate this functional on the common core, compress the resulting estimates into a 2×2 dissipation matrix, and then optimize the mixing weight α with the choice $\varpi_* = \lambda(Q_O)$.

C.1. **Preliminaries.** This appendix uses the structural estimates of Lemma 4.9 (solver bound, regularity, and perturbation control) and the notation of equations (46)–(50). We set $\varpi := \varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$b_{2,\rm eff} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{134}$$

as defined in (46). We also record the energy-admissibility threshold

$$\eta_{\rm sym}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0, \end{cases} \tag{135}$$

which is exactly the range for which the coefficient of $\|(-\mathcal{R})^{1/2}f\|^2$ in the basic energy estimate below remains nonnegative.

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}^\dagger} f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}^\dagger f_t$. In unbounded settings, by the dense-core convention from Section 3.1, for every $f_0 \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ the orbit $f_t = e^{t\mathcal{L}^\dagger} f_0$ is classical, and the algebraic identities below are first derived on the common core $\mathcal{C}$. The only model-dependent input is the existence of that core. For fixed (α, ϖ), define on $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$

$$\mathfrak{d}_{\alpha,\varpi}(f) := \Re\langle \mathcal{L}^\dagger f, f\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}^\dagger f, f\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L}^\dagger f\rangle. \tag{136}$$

Because $\mathcal{A}_\varpi$ is bounded, $\mathfrak{d}_{\alpha,\varpi}$ is continuous for the graph norm $\|f\| + \|\mathcal{L}^\dagger f\|$. Hence any quadratic-form inequality proved for $\mathfrak{d}_{\alpha,\varpi}$ on $\mathcal{C}$ extends automatically to $\mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \mathrm{dom}(\mathcal{R})$,*

$$\Re\langle \mathcal{R}f, f\rangle = \langle \mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2} f\|^2. \tag{137}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle. \tag{138}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R}f, f\rangle = -\langle (-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t denote such a classical solution. By differentiating (44) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle \partial_t f_t, f_t\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi \partial_t f_t, f_t\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \partial_t f_t\rangle \tag{139}$$

$$= \Re\langle \mathcal{L}^\dagger f_t, f_t\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}^\dagger f_t, f_t\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \mathcal{L}^\dagger f_t\rangle. \tag{140}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{141}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L}^\dagger f, f\rangle, \tag{142}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := -\alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L}^\dagger f, f\rangle - \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L}^\dagger f\rangle. \tag{143}$$

Expanding $\mathcal{L}^\dagger = \gamma\mathcal{R} - \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}^\dagger$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R}f, f\rangle - \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}}^\dagger f, f\rangle. \tag{144}$$

Using Assumption 4.5.1 ($\mathcal{R}$ coercivity) and Assumption 4.5.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R}f, f\rangle \leq -\gamma\lambda_R\|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. Since $\mathcal{V}_{\mathrm{pert}}^\dagger = -\mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$, the forward perturbation

contributes the same symmetric part as in the backward decomposition. For the symmetric part, using (34) and Cauchy–Schwarz,

$$\eta\Re\langle\mathcal{S}_{\rm pert}f,f\rangle \le \eta\|\mathcal{S}_{\rm pert}f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2}f\|\,\|f\|\Big) \tag{145}$$

$$\le \eta\Big(\big(b_2+\tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2}f\|^2\Big). \tag{146}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle\mathcal{R}f,f\rangle = \Re\langle\mathcal{R}f_F,f_F\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. If $\eta \le \eta_{\rm sym}(\gamma)$, i.e. if $\gamma - \eta a_2/2 \ge 0$, then we may use the lower bound $\|(-\mathcal{R})^{1/2}f\|^2 \ge \lambda_R\|f_F\|^2$ without changing the inequality direction. Hence

$$\gamma\Re\langle\mathcal{R}f,f\rangle + \eta\Re\langle\mathcal{S}_{\rm pert}f,f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2}f\|^2 + \eta\big(b_2+\tfrac{a_2}{2}\big)\,\|f\|^2 \tag{147}$$

$$\le -\big(\gamma\lambda_R - \eta b_{2,\rm eff}\big)\,\|f_F\|^2 + \eta\big(b_2+\tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{148}$$

where we used $\|(-\mathcal{R})^{1/2}f\|^2 \ge \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\rm eff}$ by definition. The fast mode carries the full effective coefficient $b_{2,\rm eff}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\rm eff}\big)\,\|f_F\|^2 + \eta\big(b_2+\tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{149}$$

Hypocoercive correction ($\mathcal{D}_{\rm mod}$). Recall

$$\mathcal{D}_{\rm mod}(f) = -\alpha\,\Re\langle\mathcal{A}_\varpi\mathcal{L}^\dagger f,f\rangle - \alpha\,\Re\langle\mathcal{A}_\varpi f,\mathcal{L}^\dagger f\rangle. \tag{150}$$

Using $\mathcal{L}^\dagger = \gamma\mathcal{R} - \mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert}^\dagger$ and linearity, we regroup $\mathcal{D}_{\rm mod}$ according to which part of the generator appears:

$$\mathcal{D}_{\rm mod}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{151}$$

where

$$T_{\mathcal{R}}(f) := -\alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f,f\rangle - \alpha\gamma\,\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle, \tag{152}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle\mathcal{A}_\varpi\mathcal{V}_{\rm skew}f,f\rangle + \alpha\,\Re\langle\mathcal{A}_\varpi f,\mathcal{V}_{\rm skew}f\rangle, \tag{153}$$

$$T_{\mathcal{P}}(f) := -\alpha\eta\,\Re\langle\mathcal{A}_\varpi\mathcal{V}_{\rm pert}^\dagger f,f\rangle - \alpha\eta\,\Re\langle\mathcal{A}_\varpi f,\mathcal{V}_{\rm pert}^\dagger f\rangle. \tag{154}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = -\alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f,f\rangle - \alpha\gamma\,\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle. \tag{155}$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle\mathcal{A}_\varpi f,\mathcal{R}f\rangle = \Re\langle\mathcal{R}\mathcal{A}_\varpi f,f\rangle = 0. \tag{156}$$

Moreover $\mathcal{A}_\varpi\mathcal{R}f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = -\alpha\gamma\,\Re\langle\mathcal{A}_\varpi\mathcal{R}f_F,f_S\rangle. \tag{157}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi\mathcal{R}f_F = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}(-\mathcal{R}^+)\mathcal{R}f_F = -(\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}f_F. \tag{158}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\rm skew}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\,\big|\Re\langle(\varpi - Q_O)^{-1}P_S\mathcal{V}_{\rm skew}f_F,f_S\rangle\big| = \alpha\gamma\,\big|\Re\langle f_F,\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1}f_S\rangle\big| \tag{159}$$

$$\le \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\rm skew}(\varpi - Q_O)^{-1}f_S\|. \tag{160}$$

To bound $\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \text{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$w_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{161}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\text{skew}} \psi = 0$, hence $w_\psi = \mathcal{V}_{\text{skew}} \psi = \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (29) yields

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|^2 = \|w_\psi\|^2 \le M_R \, \langle w_\psi, -\mathcal{R}^+ w_\psi \rangle. \tag{162}$$

The "extra Q_O" appears when the right-hand side is rewritten using Assumption 4.4(ii). Indeed, for $\psi \in \text{dom}(Q_O)$,

$$\langle w_\psi, -\mathcal{R}^+ w_\psi \rangle = \langle w_\psi, (-\mathcal{R}^+) w_\psi \rangle = \langle \psi, (-Q_O)\psi \rangle = \|(-Q_O)^{1/2} \psi\|^2. \tag{163}$$

Therefore

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \le \sqrt{M_R} \, \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} f_S\|. \tag{164}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \ge 0$,

$$\|(-Q_O)^{1/2} (\varpi - Q_O)^{-1}\| = \sup_{x \ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{165}$$

which implies

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \le \frac{\sqrt{M_R}}{2} \, \varpi^{-1/2} \, \|f_S\|. \tag{166}$$

Hence

$$|T_{\mathcal{R}}| \le \alpha\gamma \, \frac{\sqrt{M_R}}{2} \, \varpi^{-1/2} \, \|f_F\| \, \|f_S\|. \tag{167}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha \, \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f \rangle + \alpha \, \Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{skew}} f \rangle, \tag{168}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (36)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f \rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle \tag{169}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_F \rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_F \rangle. \tag{170}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_F \rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f \rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle. \tag{171}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\text{skew}} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{skew}} f \rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S \rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F \rangle. \tag{172}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha \, \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle + \alpha \, \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S \rangle \tag{173}$$

$$+ \alpha \, \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S \rangle + \alpha \, \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F \rangle. \tag{174}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{skew}} P_S = 0$, we have $P_F \mathcal{V}_{\text{skew}} f_S = \mathcal{V}_{\text{skew}} f_S$. On the common core, the represented identity for Q_O gives

$$\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} (-\mathcal{R}^+) \mathcal{V}_{\text{skew}} f_S = (\varpi - Q_O)^{-1} Q_O f_S. \tag{175}$$

Hence

$$\alpha \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S \rangle = -\alpha \, \langle (\varpi - Q_O)^{-1} (-Q_O) f_S, f_S \rangle. \tag{176}$$

Since $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, the spectral theorem for the non-negative self-adjoint operator $-Q_O|_{\mathcal{H}_{S,0}}$ yields

$$\alpha \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_S, f_S\rangle \le -\alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} \|f_S\|^2. \tag{177}$$

Here Assumption 4.6 identifies $\lambda(Q_O)$ as the bottom of the spectrum of $-Q_O$ on $\mathcal{H}_{S,0}$.
Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_F$ (by centering, $P_S \mathcal{V}_{\text{skew}} P_S = 0$), the third inner product vanishes: $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_S\rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$\alpha \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f_F, f_S\rangle \le \alpha \|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \|f_F\| \|f_S\| \le \alpha \left(\kappa_{\text{ell}} + \frac{\kappa_{\text{low}}}{2\sqrt{\varpi}}\right) \|f_F\| \|f_S\|. \tag{178}$$

Fast–fast remainder. Using $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ and Cauchy–Schwarz,

$$\left|\alpha \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\text{skew}} f_F\rangle\right| = \alpha \left|\Re\langle -\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi f_F, f_F\rangle\right| \le \alpha \, \|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \, \|f_F\|^2 \le \alpha \, k_{\text{reg}} \, \|f_F\|^2. \tag{179}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = -\alpha\eta \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f\rangle - \alpha\eta \Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f\rangle. \tag{180}$$

We estimate each term using Lemma 4.9 and the centered mapping property (42).
Forward term. Since $\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f \in \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f_S\rangle$. Therefore,

$$\left|\alpha\eta \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f, f\rangle\right| \le \alpha\eta \, \|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}^\dagger f\| \, \|f_S\| \le \alpha\eta \, \tilde{k}_P(\varpi) \, \|f\| \, \|f_S\| \tag{181}$$

$$\le \alpha\eta \, \tilde{k}_P(\varpi) \left(\|f_S\|^2 + \|f_F\| \, \|f_S\|\right). \tag{182}$$

Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\text{pert}} \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f\rangle = \langle \mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\left|\alpha\eta \Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}}^\dagger f\rangle\right| \le \alpha\eta \, \|\mathcal{V}_{\text{pert}} \mathcal{A}_\varpi f\| \, \|f_F\| \le \alpha\eta \, k_P(\varpi) \, \|f\| \, \|f_F\| \tag{183}$$

$$\le \alpha\eta \, k_P(\varpi) \left(\|f_F\|^2 + \|f_F\| \, \|f_S\|\right). \tag{184}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and applying Cauchy–Schwarz to terms of the form

$$|\langle X f_F, f_S\rangle| \le \|X\| \, \|f_F\| \, \|f_S\|$$

yields a closed differential inequality of the form

$$\frac{d}{dt} \mathcal{E}_{\alpha,\varpi}(f_t) \le -\mathbf{v}(t)^T \mathbf{M} \, \mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{185}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below. The fast block comes from the coercive part $\mathcal{R}$, the slow block from the DMS corrector, and the off-diagonal block from transport and perturbative couplings.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \left(\gamma\lambda_R - \eta b_{2,\text{eff}}\right) - \alpha k_{\text{reg}} - \alpha\eta k_P(\varpi) \\ &= \gamma\lambda_R - \eta b_{2,\text{eff}} - \alpha k_{\text{reg}} - \alpha\eta(b_1 + b_2) k_{\mathcal{A}} \, \varpi^{-1/2}. \end{aligned} \tag{186}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\, \tilde{k}_P(\varpi) \\ &= \alpha \frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta \left[\Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2} \right]. \end{aligned} \tag{187}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$M_{12} = -\frac{1}{2}\alpha \left[\kappa_{\text{ell}} + \left(\frac{\kappa_{\text{low}}}{2} + \gamma \frac{\sqrt{M_R}}{2} + \eta \Big(2(b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \right) \varpi^{-1/2} \right]. \tag{188}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{\det(\mathbf{M}_0)}{\operatorname{Tr}(\mathbf{M}_0)}. \tag{189}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\varpi + \lambda(Q_O))$, while the singular part of the cross term $|M_{12}^0|$ is

$$\big(\gamma\Theta + \tfrac{1}{2}\kappa_{\text{low}}\big)\varpi^{-1/2}. \tag{190}$$

A leading-order balance (maximising $\varpi(\varpi + \lambda(Q_O))^{-2}$ in the large-γ rate) motivates the choice

$$\varpi_* := \lambda(Q_O). \tag{191}$$

We do not claim here a separate global optimisation theorem over all admissible shifts; rather, ϖ_* is the canonical gap-scale choice that balances the slow coercive block against the cross term inside the determinant–trace estimate used in this paper. At this value, $\lambda(Q_O)/(\varpi_* + \lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\text{geom}}, Z(\gamma)$ from (46)–(47), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\text{reg}}, \tag{192}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{193}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\, Z(\gamma). \tag{194}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\text{reg}} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{195}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{196}$$

in agreement with (48), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\text{reg}}}. \tag{197}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\mathrm{reg}}) + \frac{1}{2}\alpha_* = \gamma\lambda_R - \alpha_*(k_{\mathrm{reg}} - \frac{1}{2})$. Since $k_{\mathrm{reg}} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\mathrm{reg}} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma\lambda_R. \tag{198}$$

Applying (189) yields

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}} =: \Lambda_{\mathrm{hypo}}(\gamma), \tag{199}$$

consistent with definition (49). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (199) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{200}$$

Thus, $\Lambda_{\mathrm{hypo}}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent coefficients K_{geom} and $2k_{\mathrm{reg}}$. These arise from the slow–fast coupling through $\mathcal{V}_{\mathrm{skew}}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\mathrm{eff}} + \alpha_*\,k_P(\varpi_*), \tag{201}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\,\tilde{k}_P(\varpi_*), \tag{202}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{203}$$

Here E_{11} contains the full effective coefficient $b_{2,\mathrm{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\mathrm{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (149)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{204}$$

Using $b_{2,\mathrm{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$\begin{aligned}\|\mathbf{E}(\gamma)\| &\leq b_{2,\mathrm{eff}} + \alpha_*(\gamma)\left[\tilde{k}_P(\varpi_*) + \tfrac{1}{2}\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big)\right] \\ &\leq b_{2,\mathrm{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\varpi_*).\end{aligned} \tag{205}$$

Recall from Lemma 4.9 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\mathrm{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2). \tag{206}$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\Big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\Big)\varpi_*^{-1/2} = C_{\text{sing}}\,\varpi_*^{-1/2},$$

so the bound (205) becomes

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\varpi_*^{-1/2}. \tag{207}$$

Substituting $\alpha_*(\gamma)$ from (48), the bound becomes

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{208}$$

consistent with definition (50).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \le \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which yields the exact matrix threshold

$$\eta^\sharp(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{209}$$

Since $\|\mathbf{E}(\gamma)\| \le B_{\text{pert}}(\gamma)$, the explicit certified threshold used in Theorem 5.1 is

$$\eta_{\text{mat}}(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \le \eta^\sharp(\gamma). \tag{210}$$

Together with the energy-admissibility condition $\eta \le \eta_{\text{sym}}(\gamma)$ coming from (149), this gives the theorem threshold

$$\eta^*(\gamma) := \min\{\eta_{\text{sym}}(\gamma), \eta_{\text{mat}}(\gamma)\}. \tag{211}$$

Substituting the explicit forms of Λ_{hypo} (49) and B_{pert} (50) yields

$$\eta_{\text{mat}}(\gamma) = \frac{\frac{1}{4}\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\text{reg}}\big)\big(2b_{2,\text{eff}} + 2\alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\big)}. \tag{212}$$

Asymptotic analysis of the certified threshold. We analyze first the scaling of $\eta_{\text{mat}}(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.

Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta_{\text{mat}}(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta_{\text{mat}}(\gamma) \asymp \mathcal{O}(1)$.

Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{213}$$

The singularity thus cancels, and at the optimal friction the matrix-controlled threshold is well-behaved:

$$\eta_{\text{mat}}(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{214}$$

Since $\eta_{\text{sym}}(\gamma_{\text{opt}}) = +\infty$ when $a_2 = 0$ and $\eta_{\text{sym}}(\gamma_{\text{opt}}) = 2\gamma_{\text{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, the overall threshold $\eta^*(\gamma_{\text{opt}}) = \min\{\eta_{\text{sym}}(\gamma_{\text{opt}}), \eta_{\text{mat}}(\gamma_{\text{opt}})\}$ has the same $\sqrt{\lambda(Q_O)}$ scaling.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{215}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\rm reg}}$, we obtain

$$\delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}}\,\frac{1}{\sqrt{\lambda_R\,\varpi_*}} = \frac{\gamma\sqrt{\lambda_R}\,\varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\rm geom}\big)^2 + 2k_{\rm reg}}. \tag{216}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta}\cdot\frac{x}{(x+K_{\rm geom})^2 + 2k_{\rm reg}}. \tag{217}$$

The elementary optimisation

$$\sup_{x>0}\frac{x}{(x+K_{\rm geom})^2 + 2k_{\rm reg}} = \frac{1}{2\big(K_{\rm geom} + \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}\big)} \tag{218}$$

together with $M_R \ge \lambda_R$ and $\sqrt{K_{\rm geom}^2 + 2k_{\rm reg}} \ge \sqrt{2k_{\rm reg}} \ge \sqrt{2}$ (the latter since $k_{\rm reg} = \sqrt{M_R/\lambda_R} \ge 1$) therefore yields

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\rm geom} + \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}\big)} \le 2^{-1/2}, \tag{219}$$

and therefore

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \le 2^{-3/2} < \frac{1}{2}. \tag{220}$$

Thus Proposition 4.11 applies with $(\alpha,\varpi) = (\alpha_*,\varpi_*)$. Hence the norm-equivalence prefactor satisfies

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}} \le 1+\sqrt{2}. \tag{221}$$

Thus, for all centered data $g \in \mathcal{H}_0$,

$$\frac{1}{2}(1-\delta_*(\gamma))\|g\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(g) \le \frac{1}{2}(1+\delta_*(\gamma))\|g\|^2. \tag{222}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\rm hypo}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{223}$$

The algebraic dissipation estimate proved above on the core is

$$\mathfrak{d}_{\alpha_*,\varpi_*}(g) \le -c(\gamma,\eta)\,\|g\|^2, \qquad g \in \mathcal{C}\cap\mathcal{H}_0. \tag{224}$$

By graph-norm continuity of $\mathfrak{d}_{\alpha_*,\varpi_*}$, the same inequality holds for every $g \in \mathrm{dom}(\mathcal{L}^\dagger)\cap\mathcal{H}_0$. Therefore, for any $g_0 \in \mathrm{dom}(\mathcal{L}^\dagger)\cap\mathcal{H}_0$ and $g_t = e^{t\mathcal{L}^\dagger}g_0$,

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(g_t) \le -c(\gamma,\eta)\,\|g_t\|^2. \tag{225}$$

Next we use the norm equivalence from Proposition 4.11 in the convenient form

$$\frac{1}{2}(1-\delta_*(\gamma))\,\|g\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(g) \le \frac{1}{2}(1+\delta_*(\gamma))\,\|g\|^2, \tag{226}$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\varpi_*}\| \in (0,1)$. The upper bound implies

$$\|g\|^2 \ge \frac{2}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(g). \tag{227}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(g_t) \le -\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(g_t). \tag{228}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(g_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\varpi_*}(g_0). \tag{229}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|g_t\|^2 \le \frac{2}{1-\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(g_t) \le \frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}t}\,\|g_0\|^2. \tag{230}$$

Therefore

$$\|g_t\| \le \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}\,e^{-\lambda_\eta t}\,\|g_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta_*(\gamma)}. \tag{231}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2+2k_{\mathrm{reg}}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}. \tag{232}$$

For the perturbation norm, we use the sharp bound (207) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\mathrm{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}}_{\alpha_*(\gamma)}\,C_{\mathrm{sing}}\,\lambda(Q_O)^{-1/2}. \tag{233}$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$\lambda_\eta \ge \frac{\frac{1}{4}\lambda_R\,\gamma}{(1+\delta_*(\gamma))(Z(\gamma)^2+2k_{\mathrm{reg}})} - \frac{\eta}{1+\delta_*(\gamma)}\left(b_{2,\mathrm{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}\,C_{\mathrm{sing}}\,\lambda(Q_O)^{-1/2}\right). \tag{234}$$

Equivalently,

$$\lambda_\eta \ge \frac{\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\big(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma)\big), \tag{235}$$

using $1+\delta_*(\gamma) \le 1+2^{-1/2}$, i.e. $(1+\delta_*(\gamma))^{-1} \ge (1+2^{-1/2})^{-1} = 2-\sqrt{2}$. This is the rigorous lower bound. The potentially singular factor $C_{\mathrm{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2+2k_{\mathrm{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate on $\mathcal{H}_0$:

$$\|e^{t\mathcal{L}^\dagger}g\|_{\mathcal{H}} \le C_*(\gamma)\,e^{-\lambda_\eta t}\|g\|_{\mathcal{H}} \le (1+\sqrt{2})\,e^{-\lambda_\eta t}\|g\|_{\mathcal{H}}. \tag{236}$$

This has been proved for $g \in \mathrm{dom}(\mathcal{L}^\dagger)\cap\mathcal{H}_0$. By Lemma 4.3, this domain is dense in $\mathcal{H}_0$. Hence for arbitrary $g \in \mathcal{H}_0$, choose $g^{(n)} \in \mathrm{dom}(\mathcal{L}^\dagger)\cap\mathcal{H}_0$ with $g^{(n)} \to g$ in $\mathcal{H}$, apply the estimate to each $g^{(n)}$, and pass to the limit using strong continuity of $e^{t\mathcal{L}^\dagger}$ and the norm equivalence above. The passage from centered data in $\mathcal{H}_0$ to physical densities/states in $\mathcal{H}_1$ is then exactly Lemma 4.2. This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine $\gamma_{\rm opt}$ maximising $\Lambda_{\rm hypo}(\gamma)$. From (199), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)} \cdot \frac{x}{(x+K_{\rm geom})^2 + 2k_{\rm reg}}. \tag{237}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\rm geom})^2 + C_0]$ with $C_0 := 2k_{\rm reg}$ is maximised when $(x+K_{\rm geom})^2 + C_0 = 2x(x+K_{\rm geom})$, equivalently $x^2 = K_{\rm geom}^2 + C_0$. Translating back:

$$\gamma_{\rm opt} = \frac{\sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\rm lift}\sqrt{\lambda(Q_O)}, \tag{238}$$

where $C_{\rm lift} := \Theta^{-1}\sqrt{K_{\rm geom}^2 + 2k_{\rm reg}}$.

Maximal rate. At $\gamma = \gamma_{\rm opt}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\rm opt} = \sqrt{K_{\rm geom}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\rm geom})^2 + C_0 = 2(K_{\rm geom}^2 + C_0 + K_{\rm geom}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\rm hypo}(\gamma_{\rm opt}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\rm geom})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{239}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped upper bound $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Certified threshold at optimal lift. The admissible perturbation range certified by Theorem 5.1 is determined as follows. From (209), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the theorem uses the explicit upper bound $B_{\rm pert}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at $\gamma_{\rm opt}$:

$$\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2 + 2k_{\rm reg}}. \tag{240}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2 + 2k_{\rm reg}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2 + 2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (207):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{241}$$

Since $b_{2,\rm eff} > 0$ by assumption and

$$B_{\rm pert}(\gamma_{\rm opt}) = b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1),$$

we have $B_{\rm pert}(\gamma_{\rm opt}) \asymp 1$. Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, the matrix-controlled threshold satisfies

$$\eta_{\rm mat}(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2B_{\rm pert}(\gamma_{\rm opt})} \asymp \sqrt{\lambda(Q_O)}. \tag{242}$$

Together with $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ when $a_2 = 0$ and $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, this yields

$$\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \eta_{\rm mat}(\gamma_{\rm opt})\} \asymp \sqrt{\lambda(Q_O)}. \tag{243}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 5.4.

D.1. **Exponential decay implies a singular value-gap bound.** By Lemma 4.3, the restriction of $e^{t\mathcal{L}^\dagger}$ to $\mathcal{H}_0$ is a C_0-semigroup with generator

$$\mathcal{L}_0^\dagger := \mathcal{L}^\dagger|_{\mathrm{dom}(\mathcal{L}^\dagger)\cap\mathcal{H}_0}.$$

For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{g\in\mathrm{dom}(T)\setminus\{0\}} \frac{\|Tg\|}{\|g\|}. \tag{244}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}^\dagger)\cap\mathcal{H}_0 = \{0\}$. *If there exist constants* $C \geq 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}^\dagger} g\| \leq C\, e^{-\nu t}\, \|g\|, \qquad \forall t \geq 0,\ \forall g \in \mathcal{H}_0, \tag{245}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}_0^\dagger$ *and*

$$\nu \leq C\, s(\mathcal{L}^\dagger). \tag{246}$$

Proof. For $\lambda > 0$, define the resolvent regularization

$$u_\lambda := -\int_0^\infty e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\, dt. \tag{247}$$

This Bochner integral is absolutely convergent because

$$\|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\| \leq C e^{-(\lambda+\nu)t}\, \|g\|, \tag{248}$$

and the right-hand side is integrable on $[0,\infty)$. Therefore,

$$\|u_\lambda\| \leq C \int_0^\infty e^{-\nu t}\, dt\, \|g\| = \frac{C}{\nu}\, \|g\|. \tag{249}$$

By the Hille–Yosida resolvent formula for the centered semigroup, $u_\lambda \in \mathrm{dom}(\mathcal{L}_0^\dagger) = \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathcal{H}_0$ and

$$(\lambda - \mathcal{L}_0^\dagger)(-u_\lambda) = g, \qquad \text{equivalently} \qquad \mathcal{L}_0^\dagger u_\lambda = g + \lambda u_\lambda. \tag{250}$$

The family $(u_\lambda)_{\lambda>0}$ is uniformly bounded in $\mathcal{H}_0$ by the estimate above and converges, as $\lambda \downarrow 0$, to

$$u := -\int_0^\infty e^{t\mathcal{L}^\dagger} g\, dt \tag{251}$$

by dominated convergence, using the integrable majorant $Ce^{-\nu t}\|g\|$. Since $\lambda u_\lambda \to 0$ in $\mathcal{H}_0$ and $\mathcal{L}_0^\dagger$ is closed, the relation $\mathcal{L}_0^\dagger u_\lambda = g + \lambda u_\lambda$ passes to the limit and gives $u \in \mathrm{dom}(\mathcal{L}_0^\dagger)$ together with $\mathcal{L}_0^\dagger u = g$. Thus $(\mathcal{L}_0^\dagger)^{-1}$ exists and satisfies $\|(\mathcal{L}_0^\dagger)^{-1}\| \leq C/\nu$, hence $s(\mathcal{L}^\dagger) = s(\mathcal{L}_0^\dagger) = \|(\mathcal{L}_0^\dagger)^{-1}\|^{-1} \geq \nu/C$, i.e. $\nu \leq C\, s(\mathcal{L}^\dagger)$. □

Corollary D.2 (Contractive semigroup bound)**.** *Under the hypotheses of Lemma D.1, assume in addition that* $e^{t\mathcal{L}^\dagger}$ *is a contraction semigroup on* $\mathcal{H}_0$. *Then*

$$\nu \leq (1 + \log C)\, s(\mathcal{L}^\dagger). \tag{252}$$

Proof. Let $T := (\log C)/\nu$ and define u_λ as in Lemma D.1. Since $e^{-\lambda t} \le 1$, contractivity on $[0,T]$ and the exponential bound on $[T,\infty)$ give

$$\|u_\lambda\| \le \int_0^T \|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\|\, dt + \int_T^\infty \|e^{-\lambda t} e^{t\mathcal{L}^\dagger} g\|\, dt \tag{253}$$

$$\le T\,\|g\| + C \int_T^\infty e^{-\nu t}\, dt\, \|g\| \tag{254}$$

$$= \frac{\log C}{\nu}\,\|g\| + \frac{Ce^{-\nu T}}{\nu}\,\|g\| \tag{255}$$

$$= \frac{1+\log C}{\nu}\,\|g\|. \tag{256}$$

Passing to $\lambda \downarrow 0$ exactly as in Lemma D.1 yields

$$\|(\mathcal{L}_0^\dagger)^{-1}\| \le \frac{1+\log C}{\nu}, \tag{257}$$

and therefore $\nu \le (1+\log C)s(\mathcal{L}^\dagger)$. □

Lemma D.3 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}^\dagger)\cap\mathcal{H}_0 = \{0\}$*. Assume moreover that* $e^{t\mathcal{L}^\dagger}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}^\dagger} f\| \le \|f\|$ *for all* $t \ge 0$ *and* $f \in \mathcal{H}_0$*. Assume there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}^\dagger} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{258}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}^\dagger) := \inf\Big\{ t \ge 0\,;\ \|e^{t\mathcal{L}^\dagger} f\| \le e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{259}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}^\dagger) \ge \frac{1}{2\, s(\mathcal{L}^\dagger)}. \tag{260}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains.*

Proof. Let $\tau > t_{\mathrm{rel}}(\mathcal{L}^\dagger)$ be such that

$$\|e^{\tau\mathcal{L}^\dagger} f\| \le e^{-1}\|f\|, \qquad \forall f \in \mathcal{H}_0. \tag{261}$$

Such a τ exists by the definition of the infimum. For any $t \ge 0$ write $t = n\tau + r$ with $n \in \mathbb{N}_0$ and $r \in [0,\tau)$. The contraction property and the semigroup identity yield

$$\|e^{t\mathcal{L}^\dagger}\| \le \|e^{r\mathcal{L}^\dagger}\|\, \|e^{\tau\mathcal{L}^\dagger}\|^n \le e^{-n}. \tag{262}$$

Since $n \ge t/\tau - 1$, this implies

$$\|e^{t\mathcal{L}^\dagger}\| \le e\, e^{-t/\tau}. \tag{263}$$

We may therefore apply Corollary D.2 with $C = e$ and $\nu = \tau^{-1}$ to obtain

$$\tau^{-1} \le (1+\log e)\, s(\mathcal{L}^\dagger) = 2\, s(\mathcal{L}^\dagger). \tag{264}$$

Hence $\tau \ge (2s(\mathcal{L}^\dagger))^{-1}$. Letting $\tau \downarrow t_{\mathrm{rel}}(\mathcal{L}^\dagger)$ gives the stated bound. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 5.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ be a normalized eigenfunction of Q_O with

$$Q_O\phi = -\lambda(Q_O)\,\phi, \qquad \|\phi\| = 1. \tag{265}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F, \tag{266}$$

and the centered test function $g_\gamma := \phi + \gamma^{-1}\psi$. By Assumption 5.3(iii), $\psi \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathrm{dom}(\mathcal{V}^\dagger_{\mathrm{pert}}) \cap \mathcal{H}_F$, hence $g_\gamma \in \mathrm{dom}(\mathcal{L}^\dagger) \cap \mathrm{dom}(\mathcal{V}^\dagger_{\mathrm{pert}}) \cap \mathcal{H}_0$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi\rangle = 0$ and therefore

$$\|g_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \geq 1. \tag{267}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi = P_F\mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{268}$$

Consequently,

$$\mathcal{L}^\dagger g_\gamma = \big(\gamma\mathcal{R} - \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}^\dagger_{\mathrm{pert}}\big)\big(\phi + \gamma^{-1}\psi\big) \tag{269}$$

$$= \underbrace{\mathcal{R}\psi - \mathcal{V}_{\mathrm{skew}}\phi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \;+\; \eta\mathcal{V}^\dagger_{\mathrm{pert}}g_\gamma. \tag{270}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}\psi + P_F\mathcal{V}_{\mathrm{skew}}\psi. \tag{271}$$

For the slow part, apply Assumption 4.4(iii) to ϕ. Since $w_\phi := P_F\mathcal{V}_{\mathrm{skew}}P_S\phi = \mathcal{V}_{\mathrm{skew}}\phi$ and $(-\mathcal{R}^+)w_\phi = -\psi$ on $\mathcal{H}_F$, we have

$$-P_S\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)w_\phi = Q_O\phi = -\lambda(Q_O)\,\phi, \tag{272}$$

and therefore $P_S\mathcal{V}_{\mathrm{skew}}\psi = \lambda(Q_O)\,\phi$. For the fast part, apply the flux regularity estimate (30) with the slow input ϕ. Using the same notation, we obtain

$$\|P_F\mathcal{V}_{\mathrm{skew}}\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)w_\phi\| \leq \kappa_{\mathrm{ell}}\,\|(-Q_O)\phi\| + \kappa_{\mathrm{low}}\,\|(-Q_O)^{1/2}\phi\| \tag{273}$$

$$= \kappa_{\mathrm{ell}}\,\lambda(Q_O) + \kappa_{\mathrm{low}}\,\sqrt{\lambda(Q_O)}. \tag{274}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\mathrm{skew}}\psi\| \leq (1 + \kappa_{\mathrm{ell}})\,\lambda(Q_O) + \kappa_{\mathrm{low}}\,\sqrt{\lambda(Q_O)}. \tag{275}$$

Hence

$$\|\mathcal{L}^\dagger g_\gamma\| \leq \frac{(1 + \kappa_{\mathrm{ell}})\,\lambda(Q_O) + \kappa_{\mathrm{low}}\sqrt{\lambda(Q_O)}}{\gamma} + \eta\,\|\mathcal{V}^\dagger_{\mathrm{pert}}g_\gamma\|. \tag{276}$$

To bound the perturbation, use Assumption 4.5.4 together with $\mathcal{V}^\dagger_{\mathrm{pert}} = -\mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$ to estimate

$$\|\mathcal{V}^\dagger_{\mathrm{pert}}f\| \leq \|\mathcal{K}_{\mathrm{pert}}f\| + \|\mathcal{S}_{\mathrm{pert}}f\| \tag{277}$$

$$\leq (a_1 + a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1 + b_2)\,\|f\|, \qquad f \in \mathrm{dom}((-\mathcal{R})^{1/2}). \tag{278}$$

Since $\phi \in \ker(\mathcal{R})$ and $g_\gamma = \phi + \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}g_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{279}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi, \psi\rangle = \langle\mathcal{V}_{\mathrm{skew}}\phi, (-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}\phi\rangle = -\langle Q_O\phi, \phi\rangle = \lambda(Q_O). \tag{280}$$

Therefore $\|(-\mathcal{R})^{1/2} g_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\mathrm{pert}}^\dagger g_\gamma\| \le (b_1 + b_2)\, \|g_\gamma\| + (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{281}$$

Dividing by $\|g_\gamma\| \ge 1$ gives

$$\frac{\|\mathcal{L}^\dagger g_\gamma\|}{\|g_\gamma\|} \le \frac{(1+\kappa_{\mathrm{ell}})\, \lambda(Q_O) + \kappa_{\mathrm{low}} \sqrt{\lambda(Q_O)}}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{282}$$

By definition of $s(\mathcal{L}^\dagger)$, this implies $s(\mathcal{L}^\dagger) \le \|\mathcal{L}^\dagger g_\gamma\|/\|g_\gamma\|$. Combining with Corollary D.2 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 6.3

We verify Assumptions 4.4 and 4.5.1–4 for the Langevin family

$$\mathcal{V}_{\mathrm{pert}} f = F(q) \cdot \nabla_p f, \tag{283}$$

with $d \ge 2$, $m \in (0, 1]$, $K \ge 0$, a confining potential $U \in C^\infty(\mathbb{R}^d)$ such that $\nabla^2 U \ge -K\,\mathrm{Id}$ and all derivatives of positive order grow at most polynomially, and a force $F \in C^\infty(\mathbb{R}^d; \mathbb{R}^d)$ whose derivatives of positive order also grow at most polynomially and which satisfies (78). We assume throughout that the position marginal $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$ has exact spectral gap $m > 0$, in the sense that m is the optimal constant in (74). The only qualitative input is the standard confining Kramers–Fokker–Planck core theory for the equilibrium part together with the polynomial-coefficient preservation of the Schwartz core by the perturbation terms; the quantitative constants are recorded in terms of $A_\nabla, B_\nabla, A_p, B_p$.

We use the Schwartz core

$$\mathcal{C}_{\mathrm{Lan}} := \mathscr{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p) \subset L^2(\mu), \tag{284}$$

where

$$d\mu(q, p) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2 + U(q))}\, dq\, dp.$$

Standard confining Kramers–Fokker–Planck theory implies that $\mathcal{C}_{\mathrm{Lan}}$ is a common invariant core for the equilibrium part $\mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R}$ and its adjoint. Because F and its derivatives have at most polynomial growth, the perturbation operators $\mathcal{V}_{\mathrm{pert}}$, $\mathcal{V}_{\mathrm{pert}}^\dagger$, $\mathcal{K}_{\mathrm{pert}}$, and $\mathcal{S}_{\mathrm{pert}}$ also preserve $\mathcal{C}_{\mathrm{Lan}}$; the same is true for P_S, P_F, and $\mathcal{R}^+$. We therefore use $\mathcal{C}_{\mathrm{Lan}}$ as the common invariant core in the sense of Section 3.1. Moreover, for $\psi \in \mathcal{C}_{\mathrm{Lan}} \cap \mathcal{H}_S = \mathscr{S}(\mathbb{R}^d)$ one has

$$\mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\mathrm{Lan}}, \qquad Q_O \psi = \beta^{-1} \Delta_q \psi - \nabla_q U \cdot \nabla_q \psi, \tag{285}$$

with $\mathcal{C}_{\mathrm{Lan}} \cap \mathcal{H}_S$ a core for the Friedrichs realization of Q_O on $L^2(\mu_q)$; see [14, Introduction and §§1–2], [8, §2], and [13, Ch. 5]. Thus Assumption 4.4 and the dense-core convention from Section 3.1 hold for this model. The remainder of the appendix is quantitative: we verify the structural constants in Assumptions 4.5.1–4 and the slow gap in Assumption 4.6. The differential identities are checked on $\mathcal{C}_{\mathrm{Lan}}$ and then extended to the relevant operator domains by the graph-norm continuity from Appendix C.

E.1. **Microscopic spectral gap, centering, and slow gap.**

The fast dissipator

$$\mathcal{R} = -p \cdot \nabla_p + \beta^{-1} \Delta_p \tag{286}$$

acts only on the Gaussian momentum variable and has spectral gap

$$\lambda_R = 1 \tag{287}$$

on $\mathcal{H}_F = \ker(\Pi_p)$. The Hamiltonian part $\mathcal{V}_{\text{skew}} = p\cdot\nabla_q - \nabla_q U\cdot\nabla_p$ is skew-adjoint in $L^2(\mu)$ because the Hamiltonian flow preserves μ. Since $\int p\, d\mu_p = 0$, one has

$$P_S\mathcal{V}_{\text{skew}}P_S = 0. \tag{288}$$

Also $P_S\mathcal{V}_{\text{pert}}P_S = 0$ because $\nabla_p\phi(q) = 0$ for slow functions.

For $\psi = \psi(q)$,

$$w_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi = p\cdot\nabla_q\psi. \tag{289}$$

Since $\mathcal{R}(p_i) = -p_i$, one has $(-\mathcal{R}^+)w_\psi = w_\psi$, so the conditioning constant is

$$M_R = 1. \tag{290}$$

The effective slow operator is

$$Q_O\psi = \beta^{-1}\Delta_q\psi - \nabla_q U\cdot\nabla_q\psi. \tag{291}$$

By the assumed spectral gap $m > 0$ for μ_q, expressed through the Poincaré inequality (74), one has

$$-\langle\psi, Q_O\psi\rangle_{L^2(\mu_q)} = \beta^{-1}\|\nabla_q\psi\|^2_{L^2(\mu_q)} \geq m\,\|\psi - \textstyle\int\psi\, d\mu_q\|^2_{L^2(\mu_q)}, \tag{292}$$

so Assumption 4.6 holds with

$$\lambda(Q_O) = m. \tag{293}$$

E.2. **Flux regularity (Assumption 4.5.3).** Define

$$Z_{ij}(p) := p_ip_j - \beta^{-1}\delta_{ij}. \tag{294}$$

Using $(-\mathcal{R}^+)w_\psi = w_\psi$, we compute

$$\begin{aligned}\mathcal{V}_{\text{skew}}w_\psi &= p\cdot\nabla_q(p\cdot\nabla_q\psi) - \nabla_q U\cdot\nabla_p(p\cdot\nabla_q\psi)\\ &= \sum_{i,j=1}^d p_ip_j\,\partial_{q_iq_j}\psi - \nabla_q U\cdot\nabla_q\psi.\end{aligned} \tag{295}$$

The p-average of this expression is

$$P_S\mathcal{V}_{\text{skew}}w_\psi = \beta^{-1}\Delta_q\psi - \nabla_q U\cdot\nabla_q\psi = Q_O\psi, \tag{296}$$

so subtracting the slow part gives

$$P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi = \sum_{i,j=1}^d (p_ip_j - \beta^{-1}\delta_{ij})\,\partial_{q_iq_j}\psi = \sum_{i,j=1}^d Z_{ij}(p)\,\partial_{q_iq_j}\psi. \tag{297}$$

The Gaussian fourth-moment identity

$$\mathbb{E}_{\mu_p}[Z_{ij}Z_{kl}] = \beta^{-2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{298}$$

therefore gives

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)w_\psi\|^2_{L^2(\mu)} = 2\beta^{-2}\,\|\nabla_q^2\psi\|^2_{L^2(\mu_q)}. \tag{299}$$

To control the Hessian, we use the weighted elliptic estimate obtained from the Bochner identity under the lower Hessian bound. Set $L := -\beta Q_O = -\Delta_q + \beta\nabla_q U\cdot\nabla_q$, the weighted Laplacian on $L^2(\mu_q)$. For every $\psi\in\mathscr{S}(\mathbb{R}^d)$, one has the commutator

$$[\partial_{q_i}, L] = \beta\,(\partial^2_{q_iq_k}U)\,\partial_{q_k}, \tag{300}$$

obtained from $[\partial_{q_i}, \nabla_q U \cdot \nabla_q] = (\partial^2_{q_i q_k} U)\partial_{q_k}$ and $[\partial_{q_i}, -\Delta_q] = 0$. Since L is self-adjoint in $L^2(\mu_q)$ and $\langle g, Lg\rangle_{L^2(\mu_q)} = \|\nabla_q g\|^2_{L^2(\mu_q)}$,

$$\begin{aligned}\|L\psi\|^2_{L^2(\mu_q)} = \langle \psi, L^2\psi\rangle_{L^2(\mu_q)} &= \sum_{i=1}^d \langle \partial_{q_i}\psi, \partial_{q_i}(L\psi)\rangle_{L^2(\mu_q)} \\ &= \sum_{i=1}^d \langle \partial_{q_i}\psi, L(\partial_{q_i}\psi)\rangle_{L^2(\mu_q)} + \beta\,\langle \nabla_q^2 U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)} \\ &= \|\nabla_q^2\psi\|^2_{L^2(\mu_q)} + \beta\,\langle \nabla_q^2 U\,\nabla_q\psi, \nabla_q\psi\rangle_{L^2(\mu_q)}. \end{aligned} \tag{301}$$

Since $\nabla_q^2 U \geq -K\,\mathrm{Id}$, the second term is bounded below by $-\beta K\|\nabla_q\psi\|^2_{L^2(\mu_q)}$, so

$$\|\nabla_q^2\psi\|^2_{L^2(\mu_q)} \leq \|L\psi\|^2_{L^2(\mu_q)} + \beta K\|\nabla_q\psi\|^2_{L^2(\mu_q)}. \tag{302}$$

Using $L = -\beta Q_O$ and

$$\beta^{-1}\|\nabla_q\psi\|^2_{L^2(\mu_q)} = \|(-Q_O)^{1/2}\psi\|^2_{L^2(\mu_q)}, \tag{303}$$

we obtain the weighted elliptic estimate

$$\|\nabla_q^2\psi\|^2_{L^2(\mu_q)} \leq \beta^2\|(-Q_O)\psi\|^2_{L^2(\mu_q)} + \beta^2 K\|(-Q_O)^{1/2}\psi\|^2_{L^2(\mu_q)}. \tag{304}$$

Combining with (299) yields

$$\|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)w_\psi\|^2_{L^2(\mu)} \leq 2\,\|(-Q_O)\psi\|^2_{L^2(\mu_q)} + 2K\,\|(-Q_O)^{1/2}\psi\|^2_{L^2(\mu_q)}, \tag{305}$$

and therefore

$$\|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)w_\psi\|_{L^2(\mu)} \leq \sqrt{2}\,\|(-Q_O)\psi\|_{L^2(\mu_q)} + \sqrt{2K}\,\|(-Q_O)^{1/2}\psi\|_{L^2(\mu_q)} \tag{306}$$

for every $\psi \in \mathscr{S}(\mathbb{R}^d)$. Since $\mathcal{C}_{\mathrm{Lan}} \cap \mathcal{H}_S = \mathscr{S}(\mathbb{R}^d)$ is a core for Q_O and the right-hand side is continuous in the graph norm of Q_O (using $\|(-Q_O)^{1/2}u\|^2 \leq \|u\|\,\|(-Q_O)u\|$), the estimate extends by density to every $\psi \in \mathrm{dom}(Q_O)$. This is exactly the two-term regularity estimate required by Assumption 4.5.3. Since $\lambda(Q_O) = m$, the example satisfies that assumption with

$$\kappa_{\mathrm{ell}} \leq \sqrt{2}, \qquad \kappa_{\mathrm{low}} \leq \sqrt{2K}. \tag{307}$$

E.3. **Perturbation bounds (Assumption 4.5.4).** The $L^2(\mu)$-adjoint of $\mathcal{V}_{\mathrm{pert}} = F(q)\cdot\nabla_p$ is

$$\mathcal{V}^\dagger_{\mathrm{pert}} = -F(q)\cdot\nabla_p + \beta F(q)\cdot p, \tag{308}$$

so the skew/symmetric decomposition is

$$\mathcal{K}_{\mathrm{pert}} = F(q)\cdot\nabla_p - \tfrac{\beta}{2}F(q)\cdot p, \qquad \mathcal{S}_{\mathrm{pert}} = \tfrac{\beta}{2}F(q)\cdot p. \tag{309}$$

On $\mathcal{C}_{\mathrm{Lan}}$, the adjoint identity above gives

$$\mathcal{K}^\dagger_{\mathrm{pert}} = -\mathcal{K}_{\mathrm{pert}}, \qquad \mathcal{S}^\dagger_{\mathrm{pert}} = \mathcal{S}_{\mathrm{pert}}. \tag{310}$$

Let $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$. Using the relative bounds (78),

$$\begin{aligned}\|\mathcal{K}_{\mathrm{pert}}f\|_{L^2(\mu)} &\leq \|F(q)\cdot\nabla_p f\|_{L^2(\mu)} + \tfrac{\beta}{2}\,\|F(q)\cdot p\,f\|_{L^2(\mu)} \\ &\leq \Big(A_\nabla + \tfrac{\beta}{2}A_p\Big)\,\|(-\mathcal{R})^{1/2}f\|_{L^2(\mu)} + \Big(B_\nabla + \tfrac{\beta}{2}B_p\Big)\,\|f\|_{L^2(\mu)}, \end{aligned} \tag{311}$$

while

$$\|\mathcal{S}_{\mathrm{pert}}f\|_{L^2(\mu)} \leq \tfrac{\beta}{2}\,\|F(q)\cdot p\,f\|_{L^2(\mu)} \leq \tfrac{\beta}{2}A_p\,\|(-\mathcal{R})^{1/2}f\|_{L^2(\mu)} + \tfrac{\beta}{2}B_p\,\|f\|_{L^2(\mu)}. \tag{312}$$

Therefore Assumption 4.5.4 holds with

$$a_1 = A_\nabla + \tfrac{\beta}{2}A_p, \qquad b_1 = B_\nabla + \tfrac{\beta}{2}B_p, \qquad a_2 = \tfrac{\beta}{2}A_p, \qquad b_2 = \tfrac{\beta}{2}B_p. \tag{313}$$

This is the entire quantitative use of the forcing hypothesis in Example 6.2: pointwise boundedness of F is not required.

Appendix F. Proof of Proposition 6.5

All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 4.5.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 4.5.2 holds since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 4.4, Assumption 4.5.3, and Assumption 4.6 are all verified explicitly. The key improvement over a single-channel coupling (e.g., $H_{\text{int}} = g\sigma_z \otimes \sigma_z$) is that the double commutator $\text{ad}^2_{\sigma_a}$ has kernel $\text{span}\{I, \sigma_a\}$; with two channels,

$$\ker(Q_O) \cap \mathcal{H}_0 = \ker(\text{ad}^2_{\sigma_x}) \cap \ker(\text{ad}^2_{\sigma_z}) \cap \{\text{tr} = 0\} = \{0\}, \tag{314}$$

so Assumption 4.6 holds with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and $\lambda(Q_O) = 4g^2/\kappa = g^2/\Gamma$.

Since all operators are bounded and $\mathcal{H}_S$ is finite-dimensional, we may take $\mathcal{C}_Q = \mathcal{H}_S$ in Assumption 4.4. The explicit formula (97) gives (24) on $\mathcal{H}_S$, and the identities (25)–(26) follow by direct block computation. For the flux regularity bound (30), define the bounded operator

$$\mathcal{K} := P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) P_F \mathcal{V}_{\text{skew}} P_S : \mathcal{H}_S \to \mathcal{H}_F. \tag{315}$$

Because $P_S \mathcal{V}_{\text{skew}} P_S = 0$, both $\mathcal{K}$ and $-Q_O$ annihilate the stationary direction $\mathbb{C}\mathbf{1}$. On the finite-dimensional mean-zero slow space $\mathcal{H}_{S,0}$, the kernel computation above shows that $-Q_O|_{\mathcal{H}_{S,0}}$ is positive definite and therefore invertible. We may thus set

$$\mathcal{T} := \mathcal{K}\,(-Q_O|_{\mathcal{H}_{S,0}})^{-1} : \mathcal{H}_{S,0} \to \mathcal{H}_F, \qquad \kappa_{\text{ell}} := \|\mathcal{T}\| < \infty. \tag{316}$$

Writing any slow vector as $\psi = c\mathbf{1} + \psi_0$ with $\psi_0 \in \mathcal{H}_{S,0}$ and using $g_{\mathbf{1}} = 0$, we obtain

$$P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi = \mathcal{K}\psi_0 = \mathcal{T}(-Q_O)\psi_0, \tag{317}$$

and hence

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\| \le \kappa_{\text{ell}}\, \|(-Q_O)\psi_0\| = \kappa_{\text{ell}}\, \|(-Q_O)\psi\|. \tag{318}$$

Thus (30) holds analytically in this finite-dimensional model in the stronger one-term form

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) w_\psi\| \le \kappa_{\text{ell}}\, \|(-Q_O)\psi\|, \tag{319}$$

i.e. one may take $\kappa_{\text{low}} = 0$. One may also take $M_R := \|\mathcal{R}\|$.

(4) For Assumption 4.5.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}^\dagger_{\text{pert}})$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}^\dagger_{\text{pert}})$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (33)–(34) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

□

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Data availability

The numerical scripts used for the illustrative computations are contained in the accompanying repository.

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